02.02.2025

1. Considerăm spaţiul vectorial real $\mathbb{R}^3$, înzestrat cu produsul scalar canonic. Pornind de la baza $\mathcal{B} = \left\{ \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$, să se construiască o bază ortonormală a lui $\mathbb{R}^3$. u_1 u_2 u_3

$v_1 = u_1 = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$

$v_2 = u_2 - \text{pr}_{v_1} u_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} \cdot v_1 = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} - \frac{-5}{13} \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 - \frac{10}{13} \\ \frac{15}{13} - 1 \end{pmatrix} = \begin{pmatrix} -1 \\ \frac{3}{13} \\ \frac{2}{13} \end{pmatrix}$

$v_3 = u_3 - \text{pr}_{v_1} u_3 - \text{pr}_{v_2} u_3 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{13} \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix} - \frac{2 + \frac{3}{13} + \frac{2}{13}}{1 + \frac{9}{169} + \frac{4}{169}} \begin{pmatrix} -1 \\ \frac{3}{13} \\ \frac{2}{13} \end{pmatrix}$

$= \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ -\frac{2}{13} \\ \frac{3}{13} \end{pmatrix} - \frac{\frac{31}{13}}{\frac{14}{13}} \begin{pmatrix} -1 \\ \frac{3}{13} \\ \frac{2}{13} \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ -2/13 \\ 3/13 \end{pmatrix} - \frac{31}{14} \begin{pmatrix} -1 \\ \frac{3}{13} \\ \frac{2}{13} \end{pmatrix}$

man wtf $= \begin{pmatrix} \frac{3}{14} \\ \frac{9}{14} \\ \frac{3}{7} \end{pmatrix}$

$\|v_1\| = \sqrt{\langle v_1, v_1 \rangle}$

$\|v_2\| = \sqrt{\langle v_2, v_2 \rangle}$

$\|v_3\| = \sqrt{\langle v_3, v_3 \rangle}$

$\Rightarrow w_1 = \frac{v_1}{\|v_1\|}$

$w_2 = \frac{v_2}{\|v_2\|}$

$w_3 = \frac{v_3}{\|v_3\|}$ calculă ...

2. Se consideră matricea $A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & 2 \\ 2 & -3 & 1 \\ 0 & 1 & 3 \end{pmatrix} \in \mathcal{M}_{4,3}(\mathbb{R})$. Folosind procedeul de ortogonalizare Gram–Schmidt, să se găsească o bază ortonormală a subspaţiului $\mathscr{C}(A)$.

$\mathscr{C}(A) = \text{sp} \left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -3 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \\ 3 \end{pmatrix} \right\}$

u_1 u_2 u_3

$\Rightarrow v_1 = u_1$

$v_2 = u_2 - p_{v_1} u_2 = u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = \begin{pmatrix}1\\-1\\-3\\1\end{pmatrix} - \frac{-6}{6}\begin{pmatrix}1\\1\\2\\0\end{pmatrix} = \begin{pmatrix}1\\-1\\-3\\1\end{pmatrix} + \begin{pmatrix}1\\1\\2\\0\end{pmatrix} = \begin{pmatrix}2\\0\\-1\\1\end{pmatrix}$

$v_3 = u_3 - p_{v_1} u_3 - p_{v_2} u_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle} v_2 = \begin{pmatrix}2\\2\\1\\3\end{pmatrix} - \frac{6}{6}\begin{pmatrix}1\\1\\2\\0\end{pmatrix} -$

$- \frac{6}{6}\begin{pmatrix}2\\0\\-1\\1\end{pmatrix} = \begin{pmatrix}2\\2\\1\\3\end{pmatrix} - \begin{pmatrix}1\\1\\2\\0\end{pmatrix} - \begin{pmatrix}2\\0\\-1\\1\end{pmatrix} = \begin{pmatrix}-1\\1\\0\\2\end{pmatrix}$

$\Rightarrow w_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\1\\2\\0\end{pmatrix}$

$w_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{6}}\begin{pmatrix}2\\0\\-1\\1\end{pmatrix}$

$w_3 = \frac{v_3}{\|v_3\|} = \frac{1}{\sqrt{6}}\begin{pmatrix}-1\\1\\0\\2\end{pmatrix}$

$\Rightarrow B = \{w_1, w_2, w_3\}$ bază ortonormată

3. Considerăm spaţiul vectorial real $\mathbb{R}_2[x]$ şi produsul scalar

$$\langle p, q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1),$$

pentru orice $p, q \in \mathbb{R}_2[x]$.

a) Folosind procedeul de ortogonalizare Gram–Schmidt, pornind de la baza $\{1, x, x^2\}$, să se găsească o bază ortonormală a lui $\mathbb{R}_2[x]$.

b) Să se găsească toate polinoamele din $\mathbb{R}_2[x]$ care sunt ortogonale pe polinomul $x^2 - 1$.

a) $u_1 = 1,\ u_2 = x,\ u_3 = x^2$

$v_1 = u_1 = 1$

$v_2 = u_2 - p_{v_1} u_2 = u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = x - \frac{-1+0+1}{3} \cdot \ldots = x$

$v_3 = u_3 - p_{v_1} u_3 - p_{v_2} u_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle} v_2 = x^2 -$

$- \frac{1+0+1}{1+0+1} \cdot 1 - \frac{-1+0+1}{2} \cdot \ldots = x^2 - 1$

$\Rightarrow w_1 = \frac{v_1}{\|v_1\|} = \frac{v_1}{\sqrt{\langle 1, 1\rangle}} = \frac{v_1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

$w_2 = \frac{v_2}{\|v_2\|} = \frac{v_2}{\sqrt{2}} = \frac{x}{\sqrt{2}}$

$w_3 = \frac{v_3}{\|v_3\|} = v_3 = x^2 - 1$

$\|x^2 - 1\| = \sqrt{\langle x^2 - 1, x^2 - 1\rangle} = \sqrt{0^2 + (-1)^2 + 0^2} = 1$

$\Rightarrow B = \left\{\frac{1}{\sqrt{2}}, \frac{x}{\sqrt{2}}, x^2 - 1\right\}$

b) $\langle p, x^2-1\rangle = 0 \quad , p \in \mathbb{R}_2[x] \Rightarrow p = ax^2+bx+c$

$$\langle p, q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1),$$

$\Rightarrow \langle p, x^2-1\rangle = (a-b+c)\cdot 0 + c(-1) + (a+b+c) = 0 \Rightarrow c = 0$

$\Rightarrow p = ax^2+bx$

4. Se consideră spaţiul vectorial real $\mathbb{R}_2[x]$, înzestrat cu produsul scalar $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\,dx$, pentru orice $p, q \in \mathbb{R}_2[x]$. Pornind de la baza $\mathcal{B} = \{1, x, x^2\}$, să se construiască o bază ortonormală a lui $\mathbb{R}_2[x]$.

$u_1 = 1 \; , u_2 = x \; , u_3 = x^2$

$v_1 = u_1 = 1$

$v_2 = u_2 - pr_{v_1} u_2 = u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = u_2 = x$

$\langle u_2, v_1\rangle = \int_{-1}^{1} x \cdot 1\,dx = \frac{x^2}{2}\Big|_{-1}^{1} = \frac{1}{2} - \frac{(-1)^2}{2} = 0$

$v_3 = u_3 - pr_{v_1} u_3 - pr_{v_2} u_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle} v_2 = x^2 - \frac{\frac{2}{3}}{2}\cdot 1 = x^2 - \frac{1}{3}$

$\langle u_3, v_1\rangle = \int_{-1}^{1} x^2\,dx = \frac{x^3}{3}\Big|_{-1}^{1} = \frac{1}{3} - \frac{(-1)}{3} = \frac{2}{3}$

$\langle v_1, v_1\rangle = \int_{-1}^{1} dx = x\Big|_{-1}^{1} = 2$

$\langle u_3, v_2\rangle = \int_{-1}^{1} x^2 \cdot x\,dx = \frac{x^4}{4}\Big|_{-1}^{1} = 0$

$\Rightarrow w_1 = \frac{v_1}{\|v_1\|} = \frac{v_1}{\sqrt{\langle v_1, v_1\rangle}} = \frac{1}{\sqrt{2}}$

$\langle v_2, v_2\rangle = \int_{-1}^{1} x^2\,dx = \frac{2}{3}$

$\langle v_3, v_3\rangle = \int_{-1}^{1} \left(x^2 - \frac{1}{3}\right)^2 dx = \int_{-1}^{1} \left(x^4 - \frac{2}{3}x^2 + \frac{1}{9}\right) dx$

$= \left(\frac{x^5}{5} - \frac{2}{3}\cdot\frac{x^3}{3} + \frac{1}{9}\cdot x\right)\Big|_{-1}^{1} = \frac{2}{5} - \frac{4}{9} + \frac{2}{9}$

$= \frac{9}{45}\cdot\frac{2}{5} - \frac{5}{45}\cdot\frac{2}{9} = \frac{8}{45}$

$w_2 = \frac{v_2}{\|v_2\|} = \frac{x}{\sqrt{\frac{2}{3}}} = \sqrt{\frac{3}{2}}\,x$

$w_3 = \frac{v_3}{\|v_3\|} = \frac{x^2 - \frac{1}{3}}{\sqrt{\frac{8}{45}}}$

5. Se consideră spaţiul vectorial real $\mathbb{R}^3$, înzestrat cu produsul scalar canonic, şi subspaţiul

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \,\middle|\, x + 2y + 2z = 0 \text{ şi } 4x - y - z = 0 \right\}.$$

Să se determine:
a) o bază şi dimensiunea pentru complementul ortogonal al lui U;
b) proiecţia ortogonală a vectorului $\mathbf{v} = (1, 4, 0)^T$ pe U.

a) $\begin{cases} x+2y+2z=0 \\ 4x-y-z=0 \quad |\cdot 2 \end{cases} \Leftrightarrow \begin{cases} x+2y+2z=0 \\ 8x-2y-2z=0 \end{cases}$ $\overset{L_1+L_2}{\Rightarrow} 9x=0 \Rightarrow x=0 \Rightarrow y=-z$

$\Rightarrow U = \left\{ \begin{pmatrix} 0 \\ -z \\ z \end{pmatrix} \,\middle|\, z \in \mathbb{R} \right\} = \operatorname{sp}\left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}$

$U^\perp = \left\{ v \in \mathbb{R}^3 \,\middle|\, \langle v, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \rangle = 0 \right\}$

$\langle v, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \rangle = 0 \Leftrightarrow v_1 \cdot 0 + v_2 \cdot (-1) + v_3 \cdot 1 = 0 \Leftrightarrow v_2 = v_3$

$\hookrightarrow v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$

$\Rightarrow U^\perp = \left\{ \begin{pmatrix} v_1 \\ v_2 \\ v_2 \end{pmatrix} \,\middle|\, v_1, v_2 \in \mathbb{R} \right\} = \operatorname{sp}\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$

$\Rightarrow \dim U^\perp = 2$

b) $\operatorname{pr}_U v = ?$

$v = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix}$

$U = \operatorname{sp}\left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}$

$u \Rightarrow w = \frac{u}{\|u\|} = \frac{u}{\sqrt{2}} = \begin{pmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \Rightarrow \mathcal{B} = \left\{ \begin{pmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \right\}$ b. ortonormată pt U

$\Rightarrow \operatorname{pr}_U v = \operatorname{pr}_w v = \frac{\langle v, w \rangle}{\langle w, w \rangle} \cdot w = \frac{-2\sqrt{2}}{1} \begin{pmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix}$

6. Se consideră spaţiul vectorial real $\mathbb{R}^3$, înzestrat cu produsul scalar canonic, şi subspaţiile

$$U_1 = \operatorname{sp}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}3\\-4\\5\end{pmatrix}\right\} \quad \text{şi} \quad U_2 = \left\{\begin{pmatrix}x\\y\\z\end{pmatrix} \in \mathbb{R}^3 \,\middle|\, x+y+z=0\right\}.$$

Să se determine o bază a complementului ortogonal al subspaţiului $U_1 \cap U_2$.

$U_1 = \operatorname{sp}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}3\\-4\\5\end{pmatrix}\right\}$

$U_2 = \left\{\begin{pmatrix}x\\y\\z\end{pmatrix} \in \mathbb{R}^3 \,\middle|\, x+y+z=0\right\}$

$x+y+z=0 \Leftrightarrow y=-x-z \Rightarrow U_2 = \left\{\begin{pmatrix}x\\-x-z\\z\end{pmatrix} \,\middle|\, x,z \in \mathbb{R}\right\} = \operatorname{sp}\left\{\begin{pmatrix}1\\-1\\0\end{pmatrix}, \begin{pmatrix}0\\-1\\1\end{pmatrix}\right\}$

$u \in U_1 \cap U_2 \Rightarrow \begin{cases} u \in U_1 \\ u \in U_2 \end{cases}$

$u \in U_1 \Rightarrow u = \alpha_1\begin{pmatrix}1\\2\\3\end{pmatrix} + \alpha_2\begin{pmatrix}3\\-4\\5\end{pmatrix}$

$u \in U_2 \Rightarrow u = \beta_1\begin{pmatrix}1\\-1\\0\end{pmatrix} + \beta_2\begin{pmatrix}0\\-1\\1\end{pmatrix}$

$\Rightarrow \alpha_1\begin{pmatrix}1\\2\\3\end{pmatrix} + \alpha_2\begin{pmatrix}3\\-4\\5\end{pmatrix} = \beta_1\begin{pmatrix}1\\-1\\0\end{pmatrix} + \beta_2\begin{pmatrix}0\\-1\\1\end{pmatrix}$

$\Leftrightarrow \begin{cases} \alpha_1 + 3\alpha_2 = \beta_1 \\ 2\alpha_1 - 4\alpha_2 = -\beta_1 - \beta_2 \\ 3\alpha_1 + 5\alpha_2 = \beta_2 \end{cases}$

$\Rightarrow 2\alpha_1 - 4\alpha_2 = -\alpha_1 - 3\alpha_2 - 3\alpha_1 - 5\alpha_2$

$\Leftrightarrow 6\alpha_1 + 4\alpha_2 = 0 \mid :2 \Leftrightarrow 3\alpha_1 + 2\alpha_2 = 0$

$\Leftrightarrow \underbrace{(3\alpha_1 + 2\alpha_2)}_{=0} + 3\alpha_2 = \beta_2 \Leftrightarrow \alpha_2 = \frac{\beta_2}{3} \Rightarrow \alpha_1 = \frac{2\frac{\beta_2}{3}}{3} = \frac{2}{9}\beta_2$

$\Rightarrow u = \frac{2}{9}\beta_2\begin{pmatrix}1\\2\\3\end{pmatrix} - \frac{1}{3}\beta_2\begin{pmatrix}3\\-4\\5\end{pmatrix} = \beta_2\begin{pmatrix}\frac{2}{9}-1\\ \frac{4}{9}+\frac{3}{4}\\ \frac{2}{3}-\frac{5}{3}\end{pmatrix} = \beta_2\begin{pmatrix}-\frac{7}{9}\\ \frac{16}{9}\\ -1\end{pmatrix}$

$\Rightarrow U_1 \cap U_2 = \operatorname{sp}\left\{\begin{pmatrix}7\\-16\\9\end{pmatrix}\right\}$

$\Rightarrow v = \frac{v}{\|v\|} = \frac{1}{\sqrt{386}}\begin{pmatrix}7\\-16\\9\end{pmatrix}$

$49 + 256 + 81 = 386$

$(U_1 \cap U_2)^\perp = \left\{ v \in \mathbb{R}^3 \,\middle|\, \langle v, w\rangle = 0 \right\}$

$\langle v, w\rangle = 0 \Leftrightarrow \left\langle \begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix}, \frac{1}{\sqrt{386}}\begin{pmatrix}7\\-16\\9\end{pmatrix} \right\rangle = \frac{1}{\sqrt{386}}(7v_1 - 16v_2 + 9v_3) = 0$

$\Rightarrow 7v_1 - 16v_2 + 9v_3 = 0 \Leftrightarrow v_2 = \frac{7v_1 + 9v_3}{16}$

$\Rightarrow (U_1 \cap U_2)^\perp = \left\{ \begin{pmatrix} v_1 \\ \frac{7v_1+9v_3}{16} \\ v_3 \end{pmatrix} \middle| v_1, v_3 \in \mathbb{R} \right\} = \mathrm{Sp}\left\{ \begin{pmatrix} 1 \\ \frac{7}{16} \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ \frac{9}{16} \\ 1 \end{pmatrix} \right\} = \mathrm{Sp}\left\{ \begin{pmatrix} 16 \\ 7 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 9 \\ 16 \end{pmatrix} \right\}$

7. a) Să se justifice şi, apoi, să se găsească descompunerea QR a matricei

$$A = \begin{pmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{pmatrix}.$$

fie $u_1 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$, $u_2 = \begin{pmatrix} 2 \\ -3 \\ 3 \end{pmatrix}$

$\Rightarrow v_1 = u_1$

$v_2 = u_2 - \mathrm{pr}_{v_1} u_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = u_2 - \frac{-11}{6} \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} =$

$= \begin{pmatrix} 2 \\ -3 \\ 3 \end{pmatrix} - \begin{pmatrix} \frac{11}{6} \\ -\frac{11}{3} \\ \frac{11}{6} \end{pmatrix} = \begin{pmatrix} \frac{1}{6} \\ \frac{2}{3} \\ \frac{1}{6} \end{pmatrix}$

$\Rightarrow w_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$

$w_2 = \frac{v_2}{\|v_2\|} = \sqrt{\frac{6}{11}} \begin{pmatrix} \frac{1}{6} \\ \frac{2}{3} \\ \frac{1}{6} \end{pmatrix}$

$\Rightarrow Q = \begin{pmatrix} \frac{-1}{\sqrt{6}} & \frac{\sqrt{\frac{6}{11}}}{6} \\ \frac{2}{\sqrt{6}} & \frac{2\sqrt{\frac{6}{11}}}{3} \\ \frac{-1}{\sqrt{6}} & \frac{\pm\sqrt{\frac{6}{11}}}{6} \end{pmatrix}$

$R = \begin{pmatrix} \langle w_1, u_1 \rangle & \langle w_1, u_2 \rangle \\ 0 & \langle w_2, u_2 \rangle \end{pmatrix} = \begin{pmatrix} \sqrt{6} & \frac{-11}{\sqrt{6}} \\ 0 & \sqrt{\frac{11}{6}} \end{pmatrix}$

b) Folosind a), să se determine soluţia în sensul celor mai mici pătrate a sistemului

$$\begin{cases} -x_1 + 2x_2 = 4 \\ 2x_1 - 3x_2 = 1 \\ -x_1 + 3x_2 = 2 \end{cases}.$$

$R\tilde{x} = Q^t b \Leftrightarrow \begin{pmatrix} \sqrt{6} & \frac{-11}{\sqrt{6}} \\ 0 & \sqrt{\frac{11}{6}} \end{pmatrix} \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} = \begin{pmatrix} \frac{-1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ \frac{\sqrt{\frac{6}{11}}}{6} & \frac{2\sqrt{\frac{6}{11}}}{3} & \frac{\pm\sqrt{\frac{6}{11}}}{6} \end{pmatrix} \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}$

$$\Rightarrow \begin{pmatrix} \sqrt{6}\,\widetilde{x}_1 - \frac{11}{\sqrt{6}}\widetilde{x}_2 \\ \sqrt{\frac{11}{6}}\,\widetilde{x}_2 \end{pmatrix} = \begin{pmatrix} \frac{-4}{\sqrt{6}} \\ \frac{\sqrt{6}}{3} \end{pmatrix}$$

$$\Rightarrow \widetilde{x}_2 = \frac{\sqrt{6}}{3}\cdot\sqrt{\frac{6}{11}} = \frac{2}{\sqrt{11}} = \frac{2\sqrt{11}}{11}$$

$$\Rightarrow \sqrt{6}\,\widetilde{x}_1 - \frac{11}{\sqrt{6}}\cdot\frac{2\sqrt{11}}{11} = \frac{-4}{\sqrt{6}} \;\Big|\cdot\sqrt{6} \iff 6\widetilde{x}_1 - 2\sqrt{11} = -4$$

$$\iff \widetilde{x}_1 = \frac{2\sqrt{11}-4}{6}$$

8. Se consideră spaţiul vectorial real $\mathbb{R}_1[x]$, înzestrat cu produsul scalar $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\,\mathrm{d}x$, pentru orice $p, q \in \mathbb{R}_1[x]$. Să se determine adjunctul operatorului liniar $T : \mathbb{R}_1[x] \to \mathbb{R}_1[x]$, cu

$$T(p) = p(0) + p(1)x, \quad \text{pentru orice } p \in \mathbb{R}_1[x].$$

$T^* = ?$ $\quad B = \{1, x\}$ baza canonică a lui $\mathbb{R}_1[x]$

$$\langle T(p), 1\rangle = \langle p(0)+p(1)x, 1\rangle = \int_{-1}^{1}(p(0)+p(1)x)\,dx$$

$p \in \mathbb{R}_1[x] \Rightarrow p = ax+b$

$$= \int_{-1}^{1}(b+(a+b)x)\,dx = \left(bx + \frac{(a+b)x^2}{2}\right)\Big|_{-1}^{1} = 2b = \langle p, T^*(1)\rangle$$

$T: \mathbb{R}_1[x] \to \mathbb{R}_1[x] \iff T^*: \mathbb{R}_1[x] \to \mathbb{R}_1[x] \Rightarrow$ Not $T^*(1) = \alpha x + \beta$

$$\Rightarrow \langle p, T^*(1)\rangle = \langle ax+b, \alpha x+\beta\rangle = \int_{-1}^{1}(ax+b)(\alpha x+\beta)\,dx = \int_{-1}^{1}(a\alpha x^2 + (a\beta+b\alpha)x + b\beta)\,dx$$

$$= \left(a\alpha\frac{x^3}{3} + (a\beta+b\alpha)\frac{x^2}{2} + b\beta x\right)\Big|_{-1}^{1} = \frac{2a\alpha}{3} + 2b\beta$$

$$\Rightarrow 2b = \frac{2}{3}a\alpha + 2b\beta \quad \forall a, b \Rightarrow \alpha = 0, \beta = 1 \Rightarrow T^*(1) = 1$$

$$\langle T(p), x\rangle = \int_{-1}^{1}(b+(a+b)x)x\,dx = \left(\frac{bx^2}{2} + (a+b)\frac{x^3}{3}\right)\Big|_{-1}^{1} = \frac{2}{3}(a+b) = \langle p, T^*(x)\rangle$$

Not $T^*(x) = \alpha x + \beta$

$$\langle p, T^*(x)\rangle = \int_{-1}^{1}(ax+b)(\alpha x+\beta)\,dx = \frac{2a\alpha}{3} + 2b\beta$$

$\Rightarrow \frac{2}{3}(a+b) = \frac{2}{3}a\alpha + 2b\beta \quad | \cdot \frac{3}{2}$

$\Leftrightarrow a+b = a\alpha + 3b\beta \quad \forall a, b \Rightarrow \alpha = 1, \beta = \frac{1}{3} \Rightarrow T^*(x) = x + \frac{1}{3}$

$\Rightarrow T^*(p) = T^*(ax+b) = a\left(x+\frac{1}{3}\right) + b \cdot 1 = ax + \frac{a}{3} + \underbrace{b}_{p(0)} = p(0) + (p(1)-p(0))\left(x+\frac{1}{3}\right)$

9. Considerăm spaţiul vectorial real $\mathbb{R}_2[x]$, înzestrat cu produsul scalar $\langle p, q\rangle = \int_{-1}^{1} p(x)\, q(x)\, \mathrm{d}x$, pentru orice $p, q \in \mathbb{R}_2[x]$, şi operatorul liniar $T : \mathbb{R}_2[x] \to \mathbb{R}_2[x]$, cu $T(p)(x) = \left(1-x^2\right) p''(x) - 2xp'(x)$, pentru orice $p \in \mathbb{R}_2[x]$. Să se arate că T este hermitian.

T hermitian $\Leftrightarrow T = T^* \Rightarrow \langle T(p), q\rangle = \langle p, T(q)\rangle$

$$\langle T(p), q\rangle = \int_{-1}^{1} \left((1-x^2)p''(x) - 2xp'(x)\right) \cdot q \, dx$$

$p, q \in \mathbb{R}_2[x] \Leftrightarrow p = ax^2+bx+c$, $q = dx^2+ex+f$

$$= \int_{-1}^{1} \left((1-x^2)\cdot 2a - 2x\cdot(2ax+b)\right)(dx^2+ex+f)\,dx$$

$$= \int_{-1}^{1} (2a - 2ax^2 - \underbrace{4ax^2}_{-6ax^2} - 2bx)(dx^2+ex+f)\,dx$$

$$= \int_{-1}^{1} (2adx^2 + 2aex + 2af - 6adx^4 - 6aex^3 - 6afx^2 - 2bdx^3 - 2bex^2 - 2bfx)\,dx$$

($2adx^2$: 0; $-6aex^3$: 0; $-2bdx^3$: 0; $-2bfx$: 0)

$= \ldots$ calcule $\Rightarrow$ sunt egale

10. Să se determine matricea $A \in \mathcal{M}_2(\mathbb{R})$ ştiind că aceasta are valorile proprii $\lambda_1 = -1$ şi $\lambda_2 = 8$ şi subspaţiile proprii corespunzătoare $V_{-1} = \mathrm{sp}\left\{(1,2)^T\right\}$ şi, respectiv, $V_8 = \mathrm{sp}\left\{(1,-1)^T\right\}$.

$P = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$, $D = \begin{pmatrix} -1 & 0 \\ 0 & 8 \end{pmatrix}$

$\hookrightarrow P^{-1} = \frac{1}{-3}\begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$

$$\Rightarrow A = PDP^{-1} = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 0 & 8 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -1 & 8 \\ -2 & -8 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 15 & -9 \\ -18 & 6 \end{pmatrix} =$$

$$= \begin{pmatrix} 5 & -3 \\ -6 & 2 \end{pmatrix}$$

04.02.2025

11. Să se cerceteze dacă există o matrice inversabilă, $P \in \mathcal{M}_3(\mathbb{R})$, şi o matrice diagonală, $D \in \mathcal{M}_3(\mathbb{R})$, astfel încât $A = PDP^{-1}$, unde:

a) $A = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$; b) $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$; c) $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$;

d) $A = \begin{pmatrix} 6 & 3 & -2 \\ -4 & -1 & 2 \\ 13 & 9 & -3 \end{pmatrix}$; e) $A = \begin{pmatrix} -7 & -9 & 9 \\ 3 & 5 & -3 \\ -3 & -3 & 5 \end{pmatrix}$.

a) $A = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$

$\det(A - \lambda I_3) = 0 \Leftrightarrow \begin{vmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{vmatrix} = (1-\lambda)(-5-\lambda)(4-\lambda) - 54 - 54 + 18(5+\lambda) + 18(1-\lambda) +$

$+ 9(4-\lambda) = (-5+4\lambda+\lambda^2)(4-\lambda) - 108 + 90 + 18\lambda +$

$+ 18 - 18\lambda + 36 - 9\lambda = -20 + 5\lambda + 16\lambda - 4\lambda^2 + 4\lambda^2 - \lambda^3$

$+ 36 - 9\lambda = -\lambda^3 + 12\lambda + 16 = 0$

$\Leftrightarrow -\lambda^3 + 16\lambda - 4\lambda + 16 = 0 \Leftrightarrow -\lambda(\lambda^2 - 16) - 4(\lambda - 4) = 0 \Leftrightarrow -\lambda(\lambda-4)(\lambda+4) - 4(\lambda-4) = 0$

$\Leftrightarrow (\lambda-4)(-\lambda^2 - 4\lambda - 4) = 0 \Leftrightarrow (\lambda-4)(\lambda+2)^2 = 0 \Rightarrow \lambda_1 = 4, \lambda_2 = -2$

I. $AV = \lambda_1 V \Leftrightarrow \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = 4 \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$

$\Leftrightarrow \begin{cases} v_1 - 3v_2 + 3v_3 = 4v_1 \\ 3v_1 - 5v_2 + 3v_3 = 4v_2 \\ 6v_1 - 6v_2 + 4v_3 = 4v_3 \end{cases} \Leftrightarrow \begin{cases} -v_1 - v_2 + v_3 = 0 \Rightarrow 2v_1 = v_3 \\ v_1 - 3v_2 + v_3 = 0 \\ v_1 - v_2 = 0 \Rightarrow v_1 = v_2 \end{cases}$

$\Rightarrow V_4 = \mathrm{Sp}\left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right\}$

II. $AV = \lambda_2 V \Leftrightarrow \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = -2 \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$

$\Leftrightarrow \begin{cases} v_1 - 3v_2 + 3v_3 = -2v_1 \\ 3v_1 - 5v_2 + 3v_3 = -2v_2 \\ 6v_1 - 6v_2 + 4v_3 = -2v_3 \end{cases} \Leftrightarrow \begin{cases} v_1 - v_2 + v_3 = 0 \\ v_1 - v_2 + v_3 = 0 \\ v_1 - v_2 + v_3 = 0 \end{cases}$ $\quad v_2 = v_1 + v_3$

$\Rightarrow V_4 = \mathrm{Sp}\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$

$\Rightarrow P = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}, D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix} \Rightarrow A = PDP^{-1}$

b) $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$

$\det(A - \lambda I_3) = 0 \Leftrightarrow \begin{vmatrix} 2-\lambda & 0 & 0 \\ 1 & 2-\lambda & 1 \\ -1 & 0 & 1-\lambda \end{vmatrix} = (2-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ 0 & 1-\lambda \end{vmatrix} = (2-\lambda)^2(1-\lambda) = 0$

$\Rightarrow \lambda_1 = 2, \lambda_2 = 1$

I. $AV = 2V$

$\Leftrightarrow \begin{cases} 2v_1 = 2v_1 \\ v_1 + 2v_2 + v_3 = 2v_2 \\ -v_1 + v_3 = 2v_3 \end{cases} \Leftrightarrow \begin{cases} v_1 + v_3 = 0 \\ -v_1 = v_3 \end{cases}$

$\Rightarrow V_2 = \left\{ \begin{pmatrix} v_1 \\ v_2 \\ -v_1 \end{pmatrix} \mid v_1, v_2 \in \mathbb{R} \right\} = sp\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$

I. $AV = V$

$\Leftrightarrow \begin{cases} 2v_1 = v_1 \\ v_1 + 2v_2 + v_3 = v_2 \\ -v_1 + v_3 = v_3 \end{cases} \Leftrightarrow \begin{cases} v_1 = 0 \\ v_2 + v_3 = 0 \end{cases} \Leftrightarrow v_2 = -v_3$

$\Rightarrow V_1 = sp\left\{ \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$

$\Rightarrow P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & -1 \end{pmatrix}, D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

toate subpct sunt la fel I assume ...

12. Se consideră matricea $M = \begin{pmatrix} -1 & -2 & 2 \\ 4 & 3 & -4 \\ 0 & -2 & 1 \end{pmatrix} \in \mathcal{M}_3(\mathbb{R})$. Să se determine:

a) valorile proprii ale matricei M şi câte o bază pentru subspaţiile proprii corespunzătoare acestora;

b) o matrice inversabilă, $P \in \mathcal{M}_3(\mathbb{R})$, şi o matrice diagonală, $D \in \mathcal{M}_3(\mathbb{R})$, astfel încât $M = PDP^{-1}$; justificaţi existenţa acestora;

c) matricea M^n, unde $n \in \mathbb{N}^*$;

d) matricea e^{tA}, unde $t \in \mathbb{R}$.

a) $\det(M - \lambda I_3) = 0 \Leftrightarrow \begin{vmatrix} -1-\lambda & -2 & 2 \\ 4 & 3-\lambda & -4 \\ 0 & -2 & 1-\lambda \end{vmatrix} = (-1-\lambda)(3-\lambda)(1-\lambda) - 16 + 8(1+\lambda) + 8(1-\lambda) = 0$

$\Leftrightarrow (-1-\lambda)(3-\lambda)(1-\lambda) - 16 + 8 + 8\lambda + 8 - 8\lambda = 0 \Rightarrow \lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 3$

I. $AV = -V$

$$\Leftrightarrow \begin{cases} -V_1 - 2V_2 + 2V_3 = -V_1 \\ 4V_1 + 3V_2 - 4V_3 = -V_2 \\ -2V_2 + V_3 = -V_3 \end{cases} \Leftrightarrow \begin{cases} V_2 = V_3 \\ V_1 + V_2 - V_3 = 0 \Rightarrow V_1 = 0 \end{cases}$$

$$\Rightarrow V_1 = \mathrm{Sp}\left\{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right\}$$

II. $AV = V$

$$\Leftrightarrow \begin{cases} -V_1 - 2V_2 + 2V_3 = V_1 \\ 4V_1 + 3V_2 - 4V_3 = V_2 \\ -2V_2 + V_3 = V_3 \end{cases} \Leftrightarrow \begin{cases} -V_1 - V_2 + V_3 = 0 \Rightarrow V_1 = V_3 \\ 2V_1 + V_2 - 2V_3 = 0 \\ -2V_2 = 0 \Rightarrow V_2 = 0 \end{cases}$$

$$\Rightarrow V_1 = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}\right\}$$

III. $AV = 3V$

$$\Leftrightarrow \begin{cases} -V_1 - 2V_2 + 2V_3 = 3V_1 \\ 4V_1 + 3V_2 - 4V_3 = 3V_2 \\ -2V_2 + V_3 = 3V_3 \end{cases} \Leftrightarrow \begin{cases} -2V_1 - V_2 + V_3 = 0 \\ V_1 = V_3 \\ -V_2 = V_3 \end{cases}$$

$$\Rightarrow V_1 = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}\right\}$$

b) $P = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \Rightarrow D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

c) $M^n = \underbrace{PDD^{-1}PDP^{-1}\dots PDP^{-1}}_{\text{mori}} = PD^nP^{-1}$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \overset{\substack{L_1+L_2 \\ -L_1+L_3}}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right) \overset{-L_2+L_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & -2 & -1 & 1 \end{array}\right)$$

$$\overset{\substack{L_3+L_2 \\ L_1+L_2}}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & -2 & -1 & 1 \end{array}\right) \overset{-L_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 2 & 1 & -1 \end{array}\right)$$

$$\Rightarrow M^n = P \cdot D^n \cdot P^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 3^n & 0 & 0 \\ 0 & (-1)^n & 0 \\ 0 & 0 & 1^n \end{pmatrix} \begin{pmatrix} -1 & -1 & 1 \\ -1 & 0 & 1 \\ 2 & 1 & -1 \end{pmatrix} =$$

$$= \begin{pmatrix} 3^n & 0 & 0 \\ -3^n & (-1)^n & 0 \\ 3^n & (-1)^n & 1 \end{pmatrix} \begin{pmatrix} -1 & -1 & 1 \\ -1 & 0 & 1 \\ 2 & 1 & -1 \end{pmatrix} = \begin{pmatrix} -3^n & -3^n & 3^n \\ 3^n + (-1)^{n+1} & -3^n & -3^n + (-1)^n \\ -3^n + (-1)^{n+1} + 2 & -3^n + 1 & 3^n + (-1)^n - 1 \end{pmatrix}$$

d) e^{tA}, $t \in \mathbb{R}$

$e^{tA} = P \cdot e^{tD} \cdot P^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} e^{3t} & 0 & 0 \\ 0 & e^{-t} & 0 \\ 0 & 0 & e^{t} \end{pmatrix} \begin{pmatrix} -1 & -1 & 1 \\ -1 & 0 & 1 \\ 2 & 1 & -1 \end{pmatrix}$ ... calculez

13. Să se determine o matrice ortogonală, $Q \in \mathcal{M}_3(\mathbb{R})$, şi o matrice diagonală, $D \in \mathcal{M}_3(\mathbb{R})$, astfel încât $A = QDQ^T$, unde:

a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$; b) $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$; c) $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$;

d) $A = \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & -2 \\ 2 & -2 & 0 \end{pmatrix}$; e) $A - \begin{pmatrix} -1 & 2 & 2 \\ 2 & 2 & -1 \\ 2 & -1 & 2 \end{pmatrix}$.

a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

$\det(A - \lambda I_2) = \begin{pmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} = 0 \Leftrightarrow (1-\lambda)^2 - 4 = 0 \Leftrightarrow 1 - 2\lambda + \lambda^2 - 4 = 0$

$\Leftrightarrow \lambda^2 - 2\lambda - 3 = 0 \Leftrightarrow \lambda^2 - 3\lambda + \lambda - 3 = 0 \Leftrightarrow (\lambda - 3)(\lambda + 1) = 0$

$\Rightarrow \lambda_1 = -1, \lambda_2 = 3$

I. $Av = -v \Leftrightarrow \begin{cases} v_1 + 2v_2 = -v_1 \\ 2v_1 + v_2 = -v_2 \end{cases} \Rightarrow v_2 = -v_1 \Rightarrow V_1 = \operatorname{Sp}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\}$

II. $Av = 3v \Leftrightarrow \begin{cases} v_1 + 2v_2 = 3v_1 \\ 2v_1 + v_2 = 3v_2 \end{cases} \Leftrightarrow v_1 = v_2 \Rightarrow V_3 = \operatorname{Sp}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$

$w_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cdot \frac{1}{\left\|\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$w_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\Rightarrow Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, D = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$

$\Rightarrow A = QDQ^t$

$\Leftrightarrow A = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

b) $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

$\det(A - \lambda I_3) = 0 \iff \begin{vmatrix} -\lambda & 1 & 0 \\ 1 & -\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} = 0 \iff -\lambda^3 + \lambda + \lambda = 0$

$\iff -\lambda^3 + 2\lambda = 0 \iff$

$\iff -\lambda(\lambda^2 - 2) = 0 \Rightarrow \lambda_1 = 0, \lambda_{2,3} = \pm\sqrt{2}$

I. $\lambda = 0 \Rightarrow AV = 0$

$\iff \begin{cases} v_2 = 0 \\ v_1 + v_3 = 0 \Rightarrow v_3 = -v_1 \\ v_2 = 0 \end{cases} \Rightarrow V_0 = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}\right\}$

II. $\lambda = \sqrt{2} \Rightarrow AV = \sqrt{2}V$

$\iff \left.\begin{cases} v_2 = \sqrt{2}v_1 \\ v_1 + v_3 = \sqrt{2}v_2 \\ v_2 = \sqrt{2}v_3 \end{cases}\right\} \Rightarrow v_1 = v_3 \Rightarrow 2v_1 = \sqrt{2}v_2 \iff v_2 = \sqrt{2}v_1$

$\Rightarrow V_{\sqrt{2}} = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}\right\}$

III. $\lambda = -\sqrt{2} \Rightarrow AV = -2V$

$\iff \left.\begin{cases} v_2 = -\sqrt{2}v_1 \\ v_1 + v_3 = -\sqrt{2}v_2 \\ v_2 = -\sqrt{2}v_3 \end{cases}\right\} \Rightarrow v_1 = v_3 \Rightarrow 2v_1 = -\sqrt{2}v_2 \iff v_2 = -\sqrt{2}v_1$

$\Rightarrow V_{\sqrt{2}} = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}\right\}$

$v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \Rightarrow w_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$

$v_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \Rightarrow w_2 \frac{v_2}{\|v_2\|} = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{2}}{2} \\ \frac{1}{2} \end{pmatrix}$

$w_3 = \begin{pmatrix} \frac{1}{2} \\ -\frac{\sqrt{2}}{2} \\ \frac{1}{2} \end{pmatrix}$

$\Rightarrow Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

$\Rightarrow D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & -\sqrt{2} \end{pmatrix}$

$\Rightarrow A = QDQ^t$ etc

14. Se consideră matricea $M = \begin{pmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{pmatrix} \in \mathcal{M}_3(\mathbb{R})$. Să se determine:

a) valorile proprii ale matricei M şi câte o bază pentru subspaţiile proprii corespunzătoare acestora;

b) o matrice ortogonală, $Q \in \mathcal{M}_3(\mathbb{R})$, şi o matrice diagonală, $D \in \mathcal{M}_3(\mathbb{R})$, astfel încât $M = QDQ^T$;

c) matricea M^n, unde $n \in \mathbb{N}^*$.

a) $\det(M-\lambda I_3)=0 \Leftrightarrow \begin{vmatrix} 3-\lambda & -2 & 4 \\ -2 & 6-\lambda & 2 \\ 4 & 2 & 3-\lambda \end{vmatrix} = 0 \Leftrightarrow (3-\lambda)^2(6-\lambda)-16-16-16(6-\lambda)-8(3-\lambda)=0$

$\Leftrightarrow (6-\lambda)(3-\lambda)^2 - 16(6-\lambda) - 8(7-\lambda)=0$

$\Leftrightarrow (6-\lambda)\left((3-\lambda)^2-4^2\right)-8(7-\lambda)=0 \Leftrightarrow (6-\lambda)(-1-\lambda)(7-\lambda)-8(7-\lambda)=0$

$\Leftrightarrow (7-\lambda)(-6-5\lambda+\lambda^2-8)=0 \Leftrightarrow (7-\lambda)(\lambda^2-5\lambda-14)=0 \Leftrightarrow (7-\lambda)(\lambda^2-7\lambda+2\lambda-14)=0$

$\Leftrightarrow (7-\lambda)^2(\lambda+2)=0 \Rightarrow \lambda_{1,2}=7,\ \lambda_3=-2$

I. $MV=7V$

$\Leftrightarrow \begin{cases} 3v_1-2v_2+4v_3=7v_1 \\ -2v_1+6v_2+2v_3=7v_2 \\ 4v_1+2v_2+3v_3=7v_3 \end{cases} \Leftrightarrow \begin{cases} -2v_1-v_2+2v_3=0 \\ -2v_1-v_2+2v_3=0 \\ 2v_1+v_2-2v_3=0 \end{cases}$

$\Rightarrow v_2 = 2v_3-2v_1$

$\Rightarrow V_7 = \left\{ \begin{pmatrix} v_1 \\ 2v_3-2v_1 \\ v_3 \end{pmatrix} \middle| v_1, v_3 \in \mathbb{R} \right\} = \mathrm{Sp}\left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} \right\}$

II. $MV=-2V$

$\Leftrightarrow \begin{cases} 3v_1-2v_2+4v_3=-2v_1 \\ -2v_1+6v_2+2v_3=-2v_2 \\ 4v_1+2v_2+3v_3=-2v_3 \end{cases} \Leftrightarrow \begin{cases} 5v_1-2v_2+4v_3=0 \\ -v_1+4v_2+v_3=0 \\ 4v_1+2v_2+5v_3=0 \end{cases} \rightarrow v_1 = 4v_2+v_3$

$\hookrightarrow 16v_2+4v_3+2v_2+5v_3=0$

$\Leftrightarrow 18v_2+9v_3=0 \Leftrightarrow v_3=-2v_2$

$\Rightarrow v_1 = 2v_2$

$\Rightarrow V_{-2} = \mathrm{Sp}\left\{ \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \right\}$

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b) $Q = \begin{pmatrix} \frac{1}{\sqrt{5}} & 0 & \frac{2}{3} \\ \frac{2}{\sqrt{5}} & \frac{-2}{\sqrt{5}} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{5}} & -\frac{2}{3} \end{pmatrix}$, $D = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

c) $M^n = QD^nQ^t$

15. Se consideră operatorul liniar $T : \mathbb{R}_2[x] \to \mathbb{R}_2[x]$, cu $T(p)(x) = x^2 p''(x) + xp'(x)$, pentru orice $p \in \mathbb{R}_2[x]$. Să se determine valorile proprii ale lui T şi subspaţiile proprii corespunzătoare.

$T(p)(x) = x^2 p''(x) + xp'(x) \quad \forall p \in \mathbb{R}_2[x]$

$p \in \mathbb{R}_2[x] \Rightarrow p = ax^2 + bx + c, \; a, b, c \in \mathbb{R}$

$\Rightarrow T(p)(x) = x^2 \cdot 2a + x(2ax + b) = 2ax^2 + 2ax^2 + bx = 4ax^2 + bx$

$T(p)(x) = \lambda p(x) \Leftrightarrow 4ax^2 + bx = \lambda(ax^2 + bx + c)$

$\Rightarrow \begin{cases} 4a = \lambda a \\ b = \lambda b \\ 0 = \lambda c \end{cases} \quad \forall a, b, c \Rightarrow \lambda_1 = 4, \lambda_2 = 1, \lambda_3 = 0$

I. $T(p)(x) = 4p(x) \Leftrightarrow 4ax^2 + bx = 4ax^2 + 4bx + 4c$

$\Rightarrow \begin{cases} b = 0 \\ c = 0 \end{cases}$

$\Rightarrow V_4 = \{ ax^2 \mid a \in \mathbb{R} \} = sp\{x^2\}$

II. $T(p)(x) = p(x) \Rightarrow 4ax^2 + bx = ax^2 + bx + c$

$\Rightarrow \begin{cases} 3a = 0 \\ c = 0 \end{cases} \Leftrightarrow \begin{cases} a = 0 \\ c = 0 \end{cases}$

$\Rightarrow V_1 = sp\{x\}$

III. $T(p)(x) = 0 \Rightarrow \begin{cases} a = 0 \\ b = 0 \end{cases}$

$\Rightarrow V_0 = sp\{1\}$

16. Fie $\mathbf{a} \in \mathcal{V}_3$, cu $\mathbf{a} \neq \mathbf{0}$, şi $f : \mathcal{V}_3 \to \mathcal{V}_3$, cu $f(\mathbf{x}) = \mathbf{a} \times \mathbf{x}$, pentru orice $\mathbf{x} \in \mathcal{V}_3$.

a) Să se arate că f este operator liniar pe spaţiul vectorial real $\mathcal{V}_3$.

b) Să se determine valorile proprii ale lui f şi subspaţiile proprii corespunzătoare.

c) Este operatorul liniar f diagonalizabil?

a) $f(\vec{x} + \vec{y}) = \vec{a} \times (\vec{x} + \vec{y}) = \vec{a} \times \vec{x} + \vec{a} \times \vec{y} = f(\vec{x}) + f(\vec{y})$

$f(\alpha \vec{x}) = \vec{a} \times (\alpha \vec{x}) = \alpha \vec{a} \times \vec{x} = \alpha f(\vec{x})$

$\Rightarrow f$ op. liniar

b) $f(\vec{x}) = \lambda\vec{x} \iff \vec{a}\times\vec{x} = \lambda\vec{x}$

$\vec{a}\times\vec{x} \perp \vec{a}$

$\vec{a}\times\vec{x} \perp \vec{x}$

$\vec{a}\times\vec{x} = 0 \iff \vec{a} \parallel \vec{x} \Rightarrow \vec{x} = \alpha\vec{a}$

$\Rightarrow V = sp\{\vec{a}\}$

c) $B_c = \{\vec{i}, \vec{j}, \vec{k}\}$

$f(\vec{x}) = \vec{a}\times\vec{x}$, $\vec{a} \in V_3 \Rightarrow \vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$

$$\Rightarrow T(\vec{i}) = \vec{a}\times\vec{i} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ 1 & 0 & 0 \end{vmatrix} = 0\vec{i} + a_3\vec{j} - a_2\vec{k}$$

$$T(\vec{j}) = \vec{a}\times\vec{j} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ 0 & 1 & 0 \end{vmatrix} = a_1\vec{k} - a_3\vec{i} = -a_3\vec{i} + 0\vec{j} + a_1\vec{k}$$

$$T(\vec{k}) = \vec{a}\times\vec{k} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ 0 & 0 & 1 \end{vmatrix} = a_2\vec{i} - a_1\vec{j} + 0\vec{k}$$

$$\Rightarrow Mat_{B_c}(T) = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix} = A$$

.... x face ca la restu idk

17. Să se determine termenul general al şirului $(x_n)_{n\geq 1}$, definit de relaţia de recurenţă $x_{n+2} = 3x_{n+1} - 2x_n$, pentru orice $n \geq 1$, cu $x_1 = 0$ şi $x_2 = 1$.

$$X_n = \begin{pmatrix} x_n \\ x_{n+1} \end{pmatrix} \Rightarrow \begin{pmatrix} x_{n+1} \\ x_{n+2} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}}_{A} \begin{pmatrix} x_n \\ x_{n+1} \end{pmatrix}$$

$$\Rightarrow \det(A - \lambda I_2) = 0 \iff \begin{vmatrix} -\lambda & 1 \\ -2 & 3-\lambda \end{vmatrix} = 0 \iff -\lambda(3-\lambda) + 2 = 0$$

$$\iff -3\lambda + \lambda^2 + 2 = 0 \iff \lambda^2 - 2\lambda - (\lambda - 2) = 0 \iff (\lambda - 1)(\lambda - 2) \Rightarrow \lambda_1 = 1, \lambda_2 = 2$$

I. $AV = V$

$\Leftrightarrow \begin{cases} v_2 = v_1 \\ -2v_1 + 3v_2 = v_2 \end{cases} \Rightarrow V_1 = Sp\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$

II. $AV = 2V$

$\Leftrightarrow \begin{cases} v_2 = 2v_1 \\ -2v_1 + 3v_2 = 2v_2 \end{cases} \Rightarrow V_2 = Sp\left\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\}$

$\Rightarrow P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, $D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$

$\hookrightarrow P^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$

$\Rightarrow A = PDP^{-1} \Rightarrow A^n = PD^nP^{-1}$

$X_n = A^n X_1 = PD^nP^{-1}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2^n \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2^n \\ 1 & 2^{n+1} \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2-2^n & 2^n-1 \\ 2-2^{n+1} & 2^{n+1}-1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2^n-1 \\ 2^{n+1}-1 \end{pmatrix} = \begin{pmatrix} x_n \\ x_{n+1} \end{pmatrix}$

$\Rightarrow x_n = 2^n - 1$

18. Fie V, un spaţiu vectorial peste corpul $\mathbb{K}$, $T : V \to V$, un operator liniar, şi $p \in \mathbb{K}[x]$. Dacă λ este o valoare proprie a lui T şi $\mathbf{v}$ este un vector propriu corespunzător acesteia, să se arate că $p(\lambda)$ este valoare proprie a lui $p(T)$ şi $\mathbf{v}$ este vector propriu corespunzător acesteia.

λ val proprie a lui T
v vector propriu corespunzător $\Rightarrow T(v) = \lambda v$

$p \in \mathbb{K}[x] \Rightarrow p = a_0 + a_1 x + \ldots + a_n x^n$

$\Rightarrow p(T)(v) = a_0 + a_1 T(v) + a_2 T^2(v) + \ldots + a_n T^n(v)$

$T^2(v) = T(T(v)) = T(\lambda v) = \lambda^2 v \xrightarrow[\text{inductie}]{\text{analog}} T^n(v) = \lambda^n v$

$\Rightarrow p(T)(v) = a_0 + a_1\lambda v + a_2\lambda^2 v + \ldots + a_n\lambda^n v = v(a_0 + a_1\lambda + \ldots + a_n\lambda)$

$= v \cdot p(\lambda) = p(\lambda)v$

$\Rightarrow p(\lambda)$ val proprie şi v vector propriu pt $p(T)$

19. Fie $A \in \mathcal{M}_n(\mathbb{C})$. Să se arate că matricele A şi A^T au acelaşi spectru.

$\det(A) = \det(A^t)$

$\det(A - \lambda I_n) = \det(A - \lambda I_n)^t = \det(A^t - \lambda I_n)$

deci din ec $\det(A - \lambda I_n) = 0$ rezulta aceleaşi valori proprii ca din ec $\det(A^t - \lambda I_n) = 0$

20. Fie $A \in \mathcal{M}_n(\mathbb{C})$ şi $\lambda_1, \lambda_2, ..., \lambda_n$, valorile proprii ale matricei A, numărate cu multiplicitatea lor. Să se arate că:

a) $\operatorname{Tr} A = \lambda_1 + \lambda_2 + ... + \lambda_n$;

b) $\det A = \lambda_1 \lambda_2 \cdot ... \cdot \lambda_n$.

a) $A = PDP^{-1}$, $D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \cdots & \cdots & \ddots & \cdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \Rightarrow \operatorname{Tr} D = \lambda_1 + \lambda_2 + \cdots \lambda_n$

$A = PDP^{-1} \mid \operatorname{Tr} \iff \operatorname{Tr}(A) = \operatorname{Tr}(PDP^{-1}) \iff \operatorname{Tr}(A) = \operatorname{Tr}(DP^{-1}P) = \operatorname{Tr}(D I_n) =$
$= \operatorname{Tr} D = \lambda_1 + \ldots + \lambda_n$

b) $A = PDP^{-1} \mid \det \iff \det(A) = \det(PDP^{-1}) = \det(P) \cdot \det(D) \cdot \det(P^{-1})$
$= \det D \cdot \det(P \cdot P^{-1}) = \det(D) \cdot \det(I_n) = \det D$
$= \lambda_1 \cdot \lambda_2 \cdot \ldots \cdot \lambda_n$

21. Folosind metoda transformărilor ortogonale, să se reducă la forma canonică următoarele forme pătratice:

a) $q(x_1, x_2) = 5x_1^2 - 6x_1x_2 + 5x_2^2$;

b) $q(x_1, x_2) = 10x_1^2 - 8x_1x_2 + 4x_2^2$;

c) $q(x_1, x_2) = 2x_1^2 - 2x_1x_2 + 2x_2^2$.

a) $q(x_1, x_2) = 5x_1^2 - 6x_1x_2 + 5x_2^2 \Rightarrow A = \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}$

$\Rightarrow \det(A - \lambda I_2) = \begin{vmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{vmatrix} = (5-\lambda)^2 - 3^2 = (5-\lambda-3)(5-\lambda+3) = (2-\lambda)(8-\lambda) = 0$

$\Rightarrow \lambda_1 = 2, \lambda_2 = 8$

I. $Av = 2v \iff \begin{cases} 5v_1 - 3v_2 = 2v_1 \\ -3v_1 + 5v_2 = 2v_2 \end{cases} \iff v_1 = v_2 \Rightarrow V_2 = \mathcal{L}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\} = \mathcal{L}\left\{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}\right\}$

II. $Av = 8v \iff \begin{cases} 5v_1 - 3v_2 = 8v_1 \\ -3v_1 + 5v_2 = 8v_2 \end{cases} \iff v_2 = -v_1 \Rightarrow V_8 = \mathcal{L}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\} = \mathcal{L}\left\{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}\right\}$

$$\Rightarrow Q=\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix},\ D=\begin{pmatrix}8 & 0\\ 0 & 2\end{pmatrix}$$

$$\Rightarrow A=QDQ^t$$

$$\begin{pmatrix}x_1\\ x_2\end{pmatrix}=Q\begin{pmatrix}y_1\\ y_2\end{pmatrix}\Leftrightarrow\begin{pmatrix}x_1\\ x_2\end{pmatrix}=\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}y_1\\ y_2\end{pmatrix}\Leftrightarrow\begin{cases}x_1=\frac{1}{\sqrt{2}}(y_1+y_2)\\ x_2=\frac{1}{\sqrt{2}}(-y_1+y_2)\end{cases}$$

$$\Rightarrow\tilde{q}(y_1,y_2)=8y_1^2+2y_2^2$$

c) $q(x_1,x_2)=2x_1^2-2x_1x_2+2x_2^2\Rightarrow A=\begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix}$

$$\Rightarrow \det(A-\lambda I_2)=\begin{vmatrix}2-\lambda & -1\\ -1 & 2-\lambda\end{vmatrix}=(2-\lambda)^2-1^2=(1-\lambda)(3-\lambda)$$

I. $Av=v$

$$\Leftrightarrow\begin{cases}2v_1-v_2=v_1\\ -v_1+2v_2=v_2\end{cases}\Leftrightarrow v_1=v_2\Rightarrow V_1=\mathrm{Sp}\left\{\begin{pmatrix}1\\ 1\end{pmatrix}\right\}=\mathrm{Sp}\left\{\begin{pmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{pmatrix}\right\}$$

II. $Av=3v$

$$\Leftrightarrow\begin{cases}2v_1-v_2=3v_1\\ -v_1+2v_2=3v_2\end{cases}\Leftrightarrow v_2=-v_1\Rightarrow V_3=\mathrm{Sp}\left\{\begin{pmatrix}1\\ -1\end{pmatrix}\right\}=\mathrm{Sp}\left\{\begin{pmatrix}\frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}}\end{pmatrix}\right\}$$

$$\Rightarrow Q=\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix},\ D=\begin{pmatrix}3 & 0\\ 0 & 1\end{pmatrix}$$

$$\Rightarrow A=QDQ^t$$

$$\begin{pmatrix}x_1\\ x_2\end{pmatrix}=Q\begin{pmatrix}y_1\\ y_2\end{pmatrix}\Leftrightarrow\begin{pmatrix}x_1\\ x_2\end{pmatrix}=\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}y_1\\ y_2\end{pmatrix}\Leftrightarrow\begin{cases}x_1=\frac{1}{\sqrt{2}}(y_1+y_2)\\ x_2=\frac{1}{\sqrt{2}}(-y_1+y_2)\end{cases}$$

$$\Rightarrow\tilde{q}(y_1,y_2)=3y_1^2+y_2^2$$

27. Să se determine soluţiile următoarelor ecuaţii diferenţiale:

1) $y' - y = ye^x$;

$\Leftrightarrow y' = ye^x + y = (e^x + 1)y$

$\Leftrightarrow \frac{dy}{dx} = (e^x+1)y \Leftrightarrow \frac{dy}{y} = (e^x+1)dx \Leftrightarrow \ln|y| = e^x + x + c$

$\Leftrightarrow |y| = e^{e^x + x + c} \Leftrightarrow y = \pm e^{e^x + x} \cdot e^c = k e^{e^x + x}$

2) $y'(x+1) = xy$;

$\Leftrightarrow y' = \frac{x}{x+1} y \Leftrightarrow \frac{dy}{dx} = \frac{x}{x+1} y \Leftrightarrow \frac{dy}{y} = \frac{x}{x+1} dx$

$\int \frac{x}{x+1} dx = \int 1 - \frac{1}{x+1} dx = x - \ln|x+1| + c$

$\Leftrightarrow \ln|y| = x - \ln(x+1) + c \Leftrightarrow y = \pm e^{x - \ln(x+1)} \cdot e^c$

$= k$

3) $y'(x+1) = y^2 + 1$;

$\Leftrightarrow y' = \frac{1}{x+1}(y^2+1) \Leftrightarrow \frac{dy}{dx} = \frac{1}{x+1}(y^2+1) \Leftrightarrow \frac{dy}{y^2+1} = \frac{dx}{x+1}$

$\operatorname{arctg} y = \ln|x+1| + c \Leftrightarrow y = \operatorname{tg}(\ln|x+1| + c) = \operatorname{tg}$

4) $y' - x^2y^2 = x^2$; $\Leftrightarrow y' = x^2(y^2+1) \Leftrightarrow \frac{dy}{dx} = x^2(y^2+1) \Leftrightarrow \frac{dy}{y^2+1} = x^2 dx$

$\Leftrightarrow \operatorname{arctg} y = \frac{x^3}{3} + c \Leftrightarrow y = \operatorname{tg}\left(\frac{x^3}{3} + c\right)$

5) $xy' \sin^2 y = (x+1)^2$; $\Leftrightarrow y' = \frac{(x+1)^2}{x} \cdot \frac{1}{\sin^2 y} \Leftrightarrow \frac{1}{\sin^2 y} dy = \frac{(x+1)^2}{x} dx$

$\Leftrightarrow -\operatorname{ctg} y = \frac{x^2}{2} + 2x + \ln|x| + c \Rightarrow y = \operatorname{arcctg}\left(-\frac{x^2}{2} - 2x - \ln|x| - c\right)$

$\int \frac{(x+1)^2}{x} dx = \int \frac{x^2+2x+1}{x} dx = \int x\,dx + \int 2\,dx + \int \frac{1}{x} dx = \frac{x^2}{2} + 2x + \ln|x| + c$

6) $(1+x)^2 y' + xy = 0$, cu $y(0) = 2$; $\Leftrightarrow y' = \frac{-x}{(x+1)^2} y \Leftrightarrow \frac{dy}{y} = \frac{-x}{(x+1)^2} dx$

$\int \frac{-x}{(x+1)^2} dx = \int \frac{1-t}{t^2} dt = \int t^{-2} dt - \int \frac{1}{t} dt = \frac{t^{-1}}{-1} - \ln|t| = \frac{-1}{t} - \ln|t| + c$

Not $x+1 = t \Rightarrow x = t-1 \Leftrightarrow dx = dt$

$\Leftrightarrow \ln|y| = -\frac{1}{x+1} - \ln|x+1| + c \Leftrightarrow y = \pm e^{-\frac{1}{x+1} - \ln|x+1|} e^{c} = k e^{-\frac{1}{x+1} - \ln|x+1|}$

$y(0) = 2 \Leftrightarrow k \cdot e^{-1} = 2 \Leftrightarrow k = 2e \Rightarrow y = 2e^{-\frac{1}{x+1} - \ln|x+1| + 1}$

7) $xy' = y^2 + 1$, cu $y(1) = 1$;

$\Leftrightarrow y' = (y^2+1)\frac{1}{x} \Leftrightarrow \frac{dy}{dx} = \frac{1}{x}(y^2+1) \Leftrightarrow \frac{1}{y^2+1}dy = \frac{1}{x}dx$

$\Leftrightarrow \operatorname{arctg} y = \ln|x| + c \Rightarrow y = \operatorname{tg}(\ln|x| + c)$

$y(1) = 1 \Leftrightarrow \operatorname{tg}(\ln|1| + c) = 1 \Leftrightarrow \operatorname{tg}(c) = 1 \Leftrightarrow c = \frac{\pi}{4}$

8) $(1-x^2)y' + x(y-a) = 0$, $a \in \mathbb{R}$;

$\Leftrightarrow y' = \frac{-x(y-a)}{1-x^2} \Leftrightarrow \frac{dy}{y-a} = \frac{-x}{1-x^2}dx$

$\frac{1}{2}\int \frac{-2x}{1-x^2}dx = \frac{1}{2}\int \frac{1}{t}dt = \frac{\ln|t|}{2} + c = \frac{\ln|1-x^2|}{2} + c$

Not $1-x^2 = t \Leftrightarrow -2x\,dx = dt$

$\Leftrightarrow \ln|y-a| = \frac{\ln|1-x^2|}{2} + c \Leftrightarrow y = a \pm e^{\frac{\ln|1-x^2|}{2} + c}$

9) $x^2y' + y = 1$;

$\Leftrightarrow y' = \frac{1}{x^2}(1-y) \Leftrightarrow \frac{dy}{1-y} = \frac{1}{x^2}dx \Leftrightarrow \ln|1-y| = \frac{-1}{x} + c$

$\Leftrightarrow 1-y = \pm e^{\frac{-1}{x}+c} \Leftrightarrow y = 1 \pm e^{\frac{-1}{x}+c}$

10) $xyy' = 1 - x^2$; $\Leftrightarrow y\,dy = \frac{(1-x^2)}{x}dx \Leftrightarrow \frac{y^2}{2} = \ln|x| - \frac{x^2}{2} + c \quad |\cdot 2$

$\Leftrightarrow y^2 = 2\ln|x| - x^2 + 2c \Leftrightarrow y = \pm\sqrt{2\ln|x| - x^2 + 2c}$

11) $y'(x^2y - y) + xy^2 + x = 0$; $\Leftrightarrow y' \cdot y(x^2-1) = -x(y^2+1) \Leftrightarrow \frac{y}{y^2+1}dy = \frac{-x}{x^2-1}dx$

$\frac{1}{2}\int \frac{2y}{y^2+1}dy = \frac{1}{2}\int \frac{1}{t}dt = \frac{1}{2}\ln|y^2+1| + c$

Not $y^2+1 = t \Rightarrow 2y\,dy = dt$

$-\frac{1}{2}\int \frac{2x}{x^2-1}dx = -\frac{1}{2}\ln|x^2-1| + c$

Not $x^2-1 = t \Leftrightarrow 2x\,dx = dt$

$\Rightarrow \frac{1}{2}\ln|y^2+1| = -\frac{1}{2}\ln|x^2-1| + c \,|\cdot 2 \Leftrightarrow \ln|y^2+1| = -\ln|x^2-1| + 2c$

$\Leftrightarrow y^2+1 = \pm e^{-\ln|x^2-1|} \cdot e^{2C} \Rightarrow y^2 = Ke^{-\ln|x^2-1|} - 1$ ($\pm e^{2C} = K$)

$\Leftrightarrow y = \pm\sqrt{Ke^{-\ln|x^2-1|} - 1}$

12) $y' + y\operatorname{tg} x = \sin 2x;$ $\Leftrightarrow y' = -\operatorname{tg} x \cdot y + \sin 2x$

y_o: $y' = -\operatorname{tg}x \cdot y \Leftrightarrow \frac{dy}{dx} = -\operatorname{tg}x \cdot y \Leftrightarrow \frac{dy}{y} = -\operatorname{tg}x\,dx$

$\Leftrightarrow \ln|y| = \ln|\cos x| + C \Leftrightarrow \ln|y| = \ln(|\cos x| \cdot e^C)$

$\Leftrightarrow y = \pm\cos x \cdot e^C \Leftrightarrow y = K\cos x$ ($\pm e^C = K$)

y_p: $y_p = K(x)\cos x \Rightarrow K'(x)\cdot\cos x - K(x)\sin x = -\operatorname{tg}x \cdot K(x)\cos x + \sin 2x$

$\Leftrightarrow K'(x)\cos x - K(x)\sin x = -\frac{\sin x}{\cos x}\cdot\cos x \cdot K(x) + \sin 2x$

$\Leftrightarrow K'(x) = \frac{\sin 2x}{\cos x} \Leftrightarrow K'(x) = \frac{2\sin x\cos x}{\cos x} = 2\sin x$

$\Leftrightarrow K(x) = \int 2\sin x\,dx \Leftrightarrow K(x) = -2\cos x + c_1 \Rightarrow$ alegem $c_1 = 0$

$\Rightarrow K(x) = -2\cos x$

$\Rightarrow y_p = -2\cos^2 x$

$\Rightarrow y = y_o + y_p = K\cos x - 2\cos^2 x$

14) $(x+1)y' - 2y = (x+1)^3;$ $\Leftrightarrow (x+1)y' = 2y + (x+1)^3 \mid :(x+1),\ x \neq -1$

$\Leftrightarrow y' = \frac{2}{x+1}y + (x+1)^2$

y_o: $y' = \frac{2}{x+1}y \Leftrightarrow \frac{dy}{y} = \frac{2}{x+1}dx \Leftrightarrow \ln|y| = 2\ln|x+1| + C$

$\Leftrightarrow \ln|y| = \ln((x+1)^2 \cdot e^C) \Rightarrow y = \pm(x+1)^2 e^C = K(x+1)^2$ ($\pm e^C = K$)

y_p: $y_p = K(x)\cdot(x+1)^2 \Rightarrow K'(x)(x+1)^2 + K(x)\cdot 2(x+1) = \frac{2}{x+1}\cdot K(x)(x+1)^2 + (x+1)^2$

$\Leftrightarrow K'(x) = 1 \Rightarrow K(x) = x + c_1$ luăm $c_1 = 0 \rightarrow K(x) = x$

$\Rightarrow y_p = x(x+1)^2$

$\Rightarrow y = y_o + y_p = K(x+1)^2 + x(x+1)^2$

15) $xy' - y = \ln x;$ $\Leftrightarrow y' = \frac{1}{x}y + \frac{\ln x}{x}$

y_o: $y' = \frac{1}{x}y \Leftrightarrow \frac{dy}{y} = \frac{dx}{x} \Leftrightarrow \ln|y| = \ln|x| + c$

$\Leftrightarrow y = \pm x \cdot e^c = k = kx$

y_p: $y_p = k(x)\cdot x \Rightarrow k'(x)\cdot x + k(x) = \frac{1}{x}\cdot k(x)\cdot x + \frac{\ln x}{x}$

$\Leftrightarrow k'(x) = \frac{\ln x}{x^2} \Rightarrow k(x) = -\frac{\ln x + 1}{x} + \underbrace{c_1}_{0} \Rightarrow k(x) = -\frac{\ln x + 1}{x}$

$\int \frac{\ln x}{x^2}dx = \int \frac{\ln x}{x}\cdot\frac{1}{x}dx = \int \frac{t}{e^t}dt = \int t\cdot e^{-t}dt = -te^{-t} + \int e^{-t}dt$

Not $\ln x = t \Leftrightarrow \frac{1}{x}dx = dt$, $\hookrightarrow x = e^t$

$\begin{cases} f' = e^{-t} \Rightarrow f = -e^{-t} \\ g = t \Rightarrow g' = 1 \end{cases}$ $\begin{cases} = -te^{-t} - e^{-t} + c_1 \\ = -\frac{\ln x}{x} - \frac{1}{x} + c_1 \\ = -\frac{\ln x + 1}{x} + c_1 \end{cases}$

$\Rightarrow y = y_o + y_p = kx - (\ln x + 1)$

16) $xy' - 2y = 2 - x,$ cu $y\left(\frac{1}{2}\right) = 0;$ $\Leftrightarrow y' = \frac{2}{x}y + \frac{2-x}{x}$

y_o: $y' = \frac{2}{x}y \Leftrightarrow \frac{dy}{y} = \frac{2dx}{x} \Leftrightarrow \ln|y| = 2\ln|x| + c$

$\Leftrightarrow y = \pm x^2 \cdot e^c = k = kx^2$

y_p: $y_p = k(x)\cdot x^2 \Rightarrow k'(x)\cdot x^2 + k(x)\cdot 2x = \frac{2}{x}\cdot k(x)\cdot x^2 + \frac{2-x}{x}$

$\Leftrightarrow k'(x) = \frac{2}{x^3} - \frac{1}{x^2} \Rightarrow k(x) = \frac{-4}{x^2} + \frac{1}{x} + \underbrace{c_1}_{0} \Rightarrow k(x) = \frac{-4}{x^2} + \frac{1}{x}$

$\Rightarrow y = y_o + y_p = kx^2 + x - 4$

$y\left(\frac{1}{2}\right) = 0 \Leftrightarrow k\cdot\frac{1}{4} + \frac{1}{2} - 4 = 0 \mid \cdot 4 \Leftrightarrow k = 14 \Rightarrow y = 14x^2 + x - 4$

17) $xy' + y = x\cos x,\ y\left(\frac{\pi}{2}\right) = 1;$

$\Leftrightarrow y' = -\frac{1}{x}y + \cos x$

y_o: $y' = -\frac{1}{x}y \Leftrightarrow \frac{dy}{y} = -\frac{dx}{x} \Leftrightarrow \ln|y| = \ln\left|\frac{1}{x}\right| + c \Leftrightarrow$

$\Leftrightarrow y = \pm\frac{1}{x}\cdot e^c = k = k\cdot\frac{1}{x}$

y_p: $y_p = K(x)\cdot\frac{1}{x}$ $\Rightarrow K'(x)\cdot\frac{1}{x} + K(x)\cdot\frac{-1}{x^2} = -\frac{1}{x}\cdot K(x)\cdot\frac{1}{x} + \cos x$

$\Rightarrow K'(x) = x\cos x \Rightarrow K(x) = x\sin x + \cos x$

$\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x + \cos x + \underbrace{C_1}_{=0}$

$f' = \cos x \Rightarrow f = \sin x$

$g = x \Rightarrow g' = 1$

$\Rightarrow y_p = \sin x + \frac{\cos x}{x}$

$\Rightarrow y = y_0 + y_p = \frac{k}{x} + \sin x + \frac{\cos x}{x}$

$y\left(\frac{\pi}{2}\right) = 1 \Rightarrow \frac{2k}{\pi} + \underbrace{\sin\frac{\pi}{2}}_{=1} + \frac{2}{\pi}\underbrace{\cos\frac{\pi}{2}}_{=0} = 1 \Leftrightarrow k = 0 \Rightarrow y = \sin x + \frac{\cos x}{x}$

18) $y' + y = e^{-x}$, cu $y(0) = 0$;
19) $xy' + 3y = -\frac{2}{x}$, $y(-1) = -3$;
20) $3y' + 4y = 5x^2 - 4$;
21) $y' + 3y = e^{3x}$;
22) $y' - 4y = e^x \cos x$;
23) $y' + y = \cos 2x + e^{-x}$;
24) $xy' = y + xy$;
25) $xy' = -2y + x^4$;

} la fel ca celelalte, sunt ec. liniare

26) $x^3y' + 3y = y^2x$;

Bernoulli: $y'(x) = a(x)y(x) + b(x)(y(x))^\alpha$
Subst $u = y^{1-\alpha}$

$y' = -\frac{3}{x^3}y + \frac{1}{x^2}y^2$

$\alpha = 2 \Rightarrow u = y^{1-\alpha} = y^{-1} \Rightarrow y = \frac{1}{u}$

$\Leftrightarrow -\frac{1}{u^2}\cdot u' = -\frac{3}{x^3}\cdot\frac{1}{u} + \frac{1}{x^2}\cdot\frac{1}{u^2} \;\Big|\cdot u^2$

$\Leftrightarrow -u' = -\frac{3}{x^3}u + \frac{1}{x^2} \;\Big|\cdot(-1) \Leftrightarrow u' = \frac{3}{x^3}u - \frac{1}{x^2}$

u_0: $u' = \frac{3}{x^3}u \Leftrightarrow \frac{du}{u} = \frac{3}{x^3}dx \Leftrightarrow \ln|u| = -\frac{3}{2}\cdot\frac{1}{x^2} + c \Leftrightarrow$

$\Leftrightarrow u = \underbrace{\pm e^{-\frac{3}{2x^2}}\cdot e^c}_{=k} = ke^{-\frac{3}{2x^2}}$

u_p: $u_p = k(x)e^{-\frac{3}{2x^2}} \Rightarrow k'(x)e^{-\frac{3}{2x^2}} + k(x)e^{-\frac{3}{2x^2}} \cdot \frac{3}{x^3} =$

$= \frac{3}{x^3} \cdot k(x) e^{-\frac{3}{2x^2}} - \frac{1}{x^2}$

$\Rightarrow k'(x) = -\frac{1}{x^2} \cdot e^{\frac{3}{2x^2}}$ ..... $\Rightarrow k(x) \Rightarrow u_p$

$\Rightarrow u = u_0 + u_p = \frac{1}{y} \Rightarrow y = \frac{1}{u}$

27) $y' - \frac{y}{x} = xy^2;$

$y' = \frac{1}{x}y + xy^2$

$\alpha = 2 \Rightarrow$ subst $u = y^{1-2} = y^{-1} \Rightarrow y = \frac{1}{u}$

$\Rightarrow -\frac{1}{u^2}u' = \frac{1}{x} \cdot \frac{1}{u} + x \cdot \frac{1}{u^2} \quad | \cdot u^2 \Leftrightarrow -u' = \frac{1}{x}u + x \quad | \cdot (-1)$

$\Leftrightarrow u' = -\frac{1}{x}u - x$

u_0: $u' = -\frac{1}{x}u \Leftrightarrow \frac{du}{u} = -\frac{1}{x}dx \Leftrightarrow \ln|u| = -\ln|x| + c$

$\Leftrightarrow u = \underbrace{\pm \frac{1}{x} e^c}_{=k} = k\frac{1}{x}$

u_p: $u_p = k(x)\frac{1}{x} \Rightarrow k'(x) \cdot \frac{1}{x} + k(x) \cdot \frac{-1}{x^2} = -\frac{1}{x} \cdot k(x) \frac{1}{x} - x$

$\Leftrightarrow k'(x) = -x^2 \Rightarrow k(x) = -\frac{x^3}{3} + \underbrace{c}_{=1} = -\frac{x^3}{3} \Rightarrow u_p = -\frac{x^2}{3}$

$\Rightarrow u = u_0 + u_p = k \cdot \frac{1}{x} - \frac{x^2}{3} = \frac{1}{y} \Rightarrow y = \frac{1}{\frac{k}{x} - \frac{x^2}{3}}$

28) $y' + \frac{y}{3} = e^x y^4;$

$y' = -\frac{y}{3} + e^x y^4$

$\alpha = 4 \Rightarrow$ subst $u = y^{1-\alpha} = y^{-3} = \frac{1}{y^3} \Rightarrow y = u^{-\frac{1}{3}} \Rightarrow y' = -\frac{1}{3} \cdot u^{-\frac{4}{3}} u'$

$\Rightarrow -\frac{1}{3}u^{-\frac{4}{3}}u' = \frac{-u^{-\frac{1}{3}}}{3} + e^x \cdot u^{-\frac{4}{3}} \quad | \cdot (-3u^{\frac{4}{3}})$

$\Leftrightarrow u' = u - 3e^x$

u_0: $u' = u \Leftrightarrow \frac{du}{u} = dx \Leftrightarrow \ln|u| = x + c \Rightarrow u = \underbrace{\pm e^x e^c}_{=k} = ke^x$

u_p: $u_p = k(x)e^x \Leftrightarrow k'(x)e^x + k(x)e^x = k(x)e^x - 3e^x \Rightarrow k'(x) = -3 \Rightarrow k(x) = -3x$

$\Rightarrow u = u_0 + u_p = ke^x - 3xe^x = \frac{1}{y^3} \Rightarrow y = \frac{1}{\sqrt[3]{e^x(k-3x)}}$

29) $xy' + y = xy^3;$

$\Leftrightarrow y' = -\frac{1}{x}y + y^3$

$\alpha = 3 \Rightarrow u = y^{1-3} = y^{-2} = \frac{1}{y^2} \Rightarrow y = u^{-\frac{1}{2}} \Rightarrow y' = -\frac{1}{2}u^{-\frac{3}{2}}u'$

$\Rightarrow -\frac{1}{2}u^{-\frac{3}{2}}u' = -\frac{1}{x}\cdot u^{-\frac{1}{2}} + u^{-\frac{3}{2}} \quad \Big| \cdot (-2u^{\frac{3}{2}})$

$\Leftrightarrow u' = \frac{2}{x}\cdot u - 2$

u_o: $u' = \frac{2}{x}u \Leftrightarrow \frac{du}{u} = \frac{2}{x}dx \Leftrightarrow \ln|u| = 2\ln|x| + c$

$\Leftrightarrow u = \underbrace{\pm e^c}_{=k} x^2 = kx^2$

u_p: $u_p = k(x)x^2 \Rightarrow k'(x)x^2 + \cancel{k(x)\,2x} = \frac{2}{x}\cdot\cancel{k(x)x^2} - 2$

$\Leftrightarrow k'(x) = -\frac{2}{x^2} \Leftrightarrow k(x) = \frac{2}{x} + \underbrace{c_1}_{=0} = \frac{2}{x} \Rightarrow u_p = 2x$

$\Rightarrow u = u_o + u_p = kx^2 + 2x = \frac{1}{y^2} \Rightarrow y = \frac{1}{\sqrt{kx^2+2x}}$

30) $y' + 2\frac{y}{x} = -x^2y^2\cos x.$

$\Leftrightarrow y' = \frac{-2}{x}y - x^2\cos x\, y^2$

$\alpha = 2 \Rightarrow u = y^{1-\alpha} = y^{-1} = \frac{1}{y} \Rightarrow y = \frac{1}{u} \Rightarrow y' = \frac{-1}{u^2}u'$

$\Leftrightarrow \frac{-1}{u^2}u' = \frac{-2}{x}\cdot\frac{1}{u} - x^2\cos x\,\frac{1}{u^2} \quad \Big| \cdot (-u^2)$

$\Leftrightarrow u' = \frac{2}{x}u + x^2\cos x$

u_o: $u' = \frac{2}{x}u \Leftrightarrow \frac{du}{u} = \frac{2}{x}dx \Leftrightarrow \ln|u| = 2\ln|x| + c$

$\Leftrightarrow u = \underbrace{\pm e^c}_{=k} x^2 = kx^2$

u_p: $u_p = k(x)x^2 \Rightarrow k'(x)x^2 + \cancel{k(x)\,2x} = \frac{2}{x}\cdot\cancel{k(x)x^2} + x^2\cos x$

$\Leftrightarrow k'(x) = \cos x \Leftrightarrow k(x) = \sin x + \underbrace{c_1}_{=0} = \sin x \Rightarrow u_p = \sin x\cdot x^2$

$\Rightarrow u = u_o + u_p = kx^2 + \sin x\cdot x^2 = \frac{1}{y}$

$\Rightarrow y = \frac{1}{kx^2 + \sin x\cdot x^2}$

31) $xy' = y + y^2$;

$\Leftrightarrow y' = \frac{1}{x}(y+y^2) \Leftrightarrow \frac{dy}{y^2+y} = \frac{1}{x}dy$

$\int \frac{1}{y(y+1)}dy = \int \frac{y+1-y}{y(y+1)}dy = \int \frac{1}{y}dy - \int \frac{1}{y+1}dy = \ln|y| - \ln|y+1|$
$= \ln\left|\frac{y}{y+1}\right|$

$\Rightarrow \ln\left|\frac{y}{y+1}\right| = \ln|x| + c \Leftrightarrow \frac{y}{y+1} = \pm e^c x \Leftrightarrow y = kx(y+1)$ ($\pm e^c = k$)

$\Leftrightarrow y = kxy + kx \Leftrightarrow y(1-kx) = kx \Leftrightarrow y = \frac{kx}{1-kx}$

32) $2y' + y\operatorname{tg} x = \frac{(4x+5)^2 y^3}{\cos x}$;

$\Leftrightarrow y' = -\frac{\operatorname{tg} x}{2}y + \frac{(4x+5)^2}{\cos x}y^3$

$\alpha = 3 \Leftrightarrow u = y^{1-\alpha} = y^{-2} \Rightarrow y = u^{-\frac{1}{2}} \Rightarrow y' = -\frac{1}{2}u^{-\frac{3}{2}}u'$

$\Rightarrow -\frac{1}{2}u^{-\frac{3}{2}}u' = -\frac{\operatorname{tg} x}{2}\cdot u^{-\frac{1}{2}} + \frac{(4x+5)^2}{\cos x}\cdot u^{-\frac{3}{2}} \quad | \cdot(-2u^{\frac{3}{2}})$

$\Leftrightarrow u' = \operatorname{tg} x \cdot u - \frac{2(4x+5)^2}{\cos x}$

u_0: $u' = \operatorname{tg} x\, u \Leftrightarrow \frac{du}{u} = \operatorname{tg} x\, dx \Leftrightarrow \ln|u| = -\ln|\cos x| + c$

$\Leftrightarrow u = \pm e^c \cdot \frac{1}{\cos x} = k\frac{1}{\cos x}$ ($\pm e^c = k$)

u_p: $u_p = k(x)\cdot\frac{1}{\cos x} \Rightarrow k'(x)\cdot\frac{1}{\cos x} + k(x)\cdot\operatorname{tg} x = k(x)\operatorname{tg} x - \frac{2(4x+5)^2}{\cos x}$

$\Leftrightarrow k'(x) = -2(4x+5)^2 \Rightarrow k(x) = -\frac{1}{6}(4x+5)^3$

$\frac{1}{4}\int 4(4x+5)^2 dx = \frac{1}{4}\int t^2 dt = \frac{1}{4}\frac{t^3}{3} + c_1 = \frac{1}{12}(4x+5)^3 + c_1$ ($c_1 = 0$)

Not $4x+5 = t \Leftrightarrow 4dx = dt$

$\Rightarrow u = u_0 + u_p = k\frac{1}{\cos x} - \frac{1}{6}(4x+5)^3 \cdot \frac{1}{\cos x} = \frac{1}{y^2} \Rightarrow y = \frac{1}{\sqrt{u}} = \dots$

33) $(y^2 - 3x^2)y' + 2xy = 0;$

$\Leftrightarrow (y^2 - 3x^2)y' = -2xy \Leftrightarrow y' = \frac{-2xy}{y^2 - 3x^2}$

34) $xy' = -y - xy^2;$

$\Leftrightarrow y' = -\frac{1}{x}y - y^2$

$\alpha = 2 \Rightarrow u = y^{1-\alpha} = y^{-1} = \frac{1}{y} \Rightarrow y = \frac{1}{u} \Rightarrow y' = -\frac{1}{u^2}u'$

$\Leftrightarrow -\frac{1}{u^2}u' = -\frac{1}{x}\cdot\frac{1}{u} - \frac{1}{u^2} \quad | \cdot(-u^2)$

$\Leftrightarrow u' = \frac{1}{x}u + 1$

$u_0: \quad u' = \frac{1}{x}u \Leftrightarrow \frac{du}{u} = \frac{dx}{x} \Leftrightarrow \ln|u| = \ln|x| + c \Leftrightarrow x = kx$

$u_p: \quad u_p = k(x)\cdot x \Rightarrow k'(x)\cdot x + \cancel{k(x)} = \cancel{\frac{1}{x}\cdot k(x)\,x} + 1$

$\Leftrightarrow k'(x) = \frac{1}{x} \Leftrightarrow k(x) = \ln|x| + c_1 = \ln|x|$ (we may take $c_1 = 0$)

$\Rightarrow u = u_0 + u_p = kx + \ln|x|x = \frac{1}{y} \Rightarrow y = \frac{1}{kx + \ln|x|\cdot x}$

35) $(2x - x^2y)y' = -y;$

$\Leftrightarrow y' = \frac{-y}{2x - x^2y}$

36) $xy' = -2y(1 - xy).$

$\Leftrightarrow xy' = -2y + 2xy^2 \Leftrightarrow y' = -\frac{2}{x}y + 2y^2$

$\alpha = 2 \Rightarrow u = y^{1-\alpha} = y^{-1} = \frac{1}{y} \Rightarrow y = \frac{1}{u} \Rightarrow y' = -\frac{1}{u^2}u'$

$\Leftrightarrow -\frac{1}{u^2}u' = -\frac{2}{x}\cdot\frac{1}{u} + 2\cdot\frac{1}{u^2} \quad | \cdot(-u^2)$

$\Leftrightarrow u' = \frac{2}{x}u - 2$

$u_0: \quad u' = \frac{2}{x}u \Leftrightarrow \frac{du}{u} = \frac{2}{x}dx \Leftrightarrow \ln|u| = 2\ln|x| + c \Leftrightarrow$

$\Leftrightarrow u = \pm x^2 e^c = kx^2$

u_p: $u_p = K(x)x^2 \Rightarrow K'(x)x^2 + K(x)\cdot 2x = \frac{2}{x}\cdot K(x)\cdot x^2 - 2$

$\Leftrightarrow K'(x) = -\frac{2}{x^2} \Rightarrow K(x) = \frac{2}{x} + \underbrace{c_1}_{0} = \frac{2}{x} \Rightarrow u_p = 2x$

$\Rightarrow u = u_0 + u_p = Kx^2 + 2x = \frac{1}{y} \Rightarrow y = \frac{1}{u} = \frac{1}{Kx^2+2x}$

05.02.2025

28. Să se determine soluţiile următoarelor ecuaţii diferenţiale:

a) $y' = (y-x)^2 + 1$, cu $y_p = x$ şi $y(0) = \frac{1}{2}$;

Riccati: $y'(x) = a(x)(y(x))^2 + b(x)y(x) + c(x)$
subst $y = u + y_p$

subst $y = u + y_p = u + x \Rightarrow y' = u' + 1$

$\Leftrightarrow u' + 1 = (u+x-x)^2 + 1 \Leftrightarrow u' = u^2 \Leftrightarrow \frac{du}{u^2} = dx$

$\Leftrightarrow -\frac{1}{u} = x + c \Leftrightarrow u = -\frac{1}{x+c} \Leftrightarrow y = -\frac{1}{x+c} + x$

dar $y(0) = \frac{1}{2} \Leftrightarrow -\frac{1}{c} = \frac{1}{2} \Leftrightarrow c = -2 \Leftrightarrow y = x - \frac{1}{x-2}$

b) $y' = -\frac{1}{3}y^2 - \frac{2}{3x^2}$, cu $y_p = \frac{1}{x}$ şi $y(1) = 2$, pentru $x > 0$;

subst $y = u + y_p = u + \frac{1}{x} \Rightarrow y' = u' - \frac{1}{x^2}$

$\Leftrightarrow u' - \frac{1}{x^2} = -\frac{1}{3}\left(u + \frac{1}{x}\right)^2 - \frac{2}{3x^2} \Leftrightarrow u' - \frac{1}{x^2} = -\frac{1}{3}\left(u^2 + 2\frac{u}{x} + \frac{1}{x^2}\right) - \frac{2}{3x^2}$

$\Leftrightarrow u' - \frac{1}{x^2} = -\frac{1}{3}u^2 - \frac{2}{3x}u - \underbrace{\frac{1}{3x^2} - \frac{2}{3x^2}}_{-\frac{1}{x^2}} \Leftrightarrow u' = -\frac{1}{3}u^2 - \frac{2}{3x}u$

ec Bernoulli

$\alpha = 2 \Rightarrow$ Not $v = u^{1-\alpha} = u^{-1} = \frac{1}{u} \Rightarrow u = \frac{1}{v} \Rightarrow u' = -\frac{1}{v^2}v'$

$\Leftrightarrow -\frac{1}{v^2}v' = -\frac{1}{3}\cdot\frac{1}{v^2} - \frac{2}{3x}\cdot\frac{1}{v} \Big| \cdot(-v^2) \Leftrightarrow v' = \frac{1}{3} + \frac{2}{3x}v$

v_0: $v' = \frac{2}{3x}v \Leftrightarrow \frac{dv}{v} = \frac{2}{3}\frac{dx}{x} \Leftrightarrow \ln|v| = \frac{2}{3}\ln|x| + c \Leftrightarrow$

$\Leftrightarrow v = \underbrace{\pm x^{\frac{2}{3}} e^{c}}_{=k} = kx^{\frac{2}{3}}$

v_p: $v_p = K(x)x^{\frac{2}{3}} \Rightarrow K'(x)\cdot x^{\frac{2}{3}} + K(x)\cdot\frac{2}{3}x^{-\frac{1}{3}} = \frac{2}{3x}\cdot K(x)\cdot x^{\frac{2}{3}} + \frac{1}{3}$

$\Leftrightarrow K'(x) = \frac{1}{3}\cdot x^{-\frac{2}{3}} \Leftrightarrow K(x) = \frac{1}{3}\cdot\frac{x^{\frac{1}{3}}}{\frac{1}{3}} = x^{\frac{1}{3}} \Rightarrow v_p = x$

$\Rightarrow v = v_0 + v_p = kx^{\frac{2}{3}} + x \Rightarrow u = \frac{1}{kx^{\frac{2}{3}} + x} \Rightarrow y = \frac{1}{kx^{\frac{2}{3}} + x} + \frac{1}{x}$

$y(1) = 2 \Leftrightarrow \frac{1}{k+1} + 1 = 2 \Leftrightarrow \frac{1}{k+1} = 1 \Rightarrow k = 0 \Rightarrow y = \frac{2}{x}$

c) $y'e^{-x} + y^2 - 2ye^x = 1 - e^{2x}$, cu $y_p = e^x$ şi $y(0) = 2$;

$y'e^{-x} = -y^2 + 2ye^x + 1 - e^{2x}$

$\Leftrightarrow e^{-x}y' = -(y^2 - 2e^x y + e^{2x}) + 1 \quad | \cdot e^x$

$\Leftrightarrow y' = -e^x(y - e^x)^2 + e^x$

subst $y = u + y_p = u + e^x \Rightarrow y' = u' + e^x$

$\Leftrightarrow u' + e^x = -e^x(u + e^x - e^x)^2 + e^x$

$\Leftrightarrow u' = -e^x u^2 \Leftrightarrow \frac{du}{u^2} = -e^x dx \Leftrightarrow -\frac{1}{u} = -e^x + c$

$\Leftrightarrow u = \frac{1}{e^x - c} \Rightarrow y = \frac{1}{e^x - c} + e^x$

dar $y(0) = 2 \Leftrightarrow \frac{1}{1-c} + 1 = 2 \Leftrightarrow c = 0 \Rightarrow y = \frac{1}{e^x} + e^x$

d) $y' = y^2 - \frac{y}{x} - \frac{1}{x^2}$, cu $y_p = \frac{1}{x}$ şi $y(1) = 2$, pentru $x > 0$.

subst $y = u + y_p = u + \frac{1}{x} \Rightarrow y' = u' - \frac{1}{x^2}$

$\Leftrightarrow u' - \frac{1}{x^2} = \left(u + \frac{1}{x}\right)^2 - \left(u + \frac{1}{x}\right)\cdot\frac{1}{x} - \frac{1}{x^2}$

$\Leftrightarrow u' = u^2 + \frac{u}{x} + \frac{1}{x^2} - \frac{u}{x} - \frac{1}{x^2}$

$\Leftrightarrow u' = \frac{1}{x}u + u^2$ ec Bernoulli

$\Rightarrow \alpha = 2 \Rightarrow v = u^{1-\alpha} = u^{-1} = \frac{1}{u} \Rightarrow u = \frac{1}{v} \Rightarrow u' = -\frac{1}{v^2}v'$

$\Leftrightarrow -\frac{1}{v^2}\cdot v' = \frac{1}{x}\cdot\frac{1}{v} + \frac{1}{v^2} \quad |\cdot(-v^2) \Leftrightarrow v' = -\frac{1}{x}v - 1$

v_o: $v' = -\frac{1}{x}v \Leftrightarrow \frac{dv}{v} = -\frac{1}{x}dx \Leftrightarrow \ln|v| = -\ln|x| + c$

$\Leftrightarrow v = \pm x^{-1}\cdot e^c = kx^{-1}$ ($\pm e^c = k$)

v_p: $v_p = k(x)\cdot x^{-1} \Leftrightarrow k'(x)\cdot x^{-1} + k(x)(-1)x^{-2} = -\frac{1}{x}k(x)\cdot x^{-1} - 1$

$\Leftrightarrow k'(x) = -x \Leftrightarrow k(x) = -\frac{x^2}{2} + c_1 = k(x) \Rightarrow v_p = -\frac{x^2}{2}\cdot\frac{1}{x} = -\frac{x}{2}$ ($c_1 = 0$)

$\Rightarrow v = v_o + v_p = kx^{-2} - \frac{x}{2}$

dar $u = \frac{1}{v} \Rightarrow u = \frac{1}{kx^{-1} - \frac{x}{2}} \Rightarrow y = u + y_p = \frac{1}{kx^{-1} - \frac{x}{2}} + \frac{1}{x}$

dar $y(1) = 2 \iff \frac{1}{k - \frac{1}{2}} + 1 = 2 \iff \frac{1}{k - \frac{1}{2}} = 1 \iff k - \frac{1}{2} = 1 \iff k = \frac{3}{2}$

$$\Rightarrow y = \frac{2}{3x^{-2} - x} + \frac{1}{x}$$

29. Aflaţi soluţiile următoarelor ecuaţii diferenţiale liniare cu coeficienţi constanţi:

a) $y'' + y' - 2y = e^{2x}$;

I. $y'' + y' - 2y = 0$

$\lambda^2 + \lambda - 2 = 0 \iff \lambda^2 + 2\lambda - (\lambda + 2) = 0 \iff (\lambda - 1)(\lambda + 2) = 0 \Rightarrow \lambda_1 = -2, \lambda_2 = 1$

$\Rightarrow e^x$ şi e^{-2x} sol

$\Rightarrow y_0 = c_1 e^x + c_2 e^{-2x}$

II. $e^{2x} = e^{ax}(Q(x)\cos(bx) + R(x)\sin(bx)) \Rightarrow \begin{cases} a = 2 \\ b = 0 \\ Q(x) = 1 \Rightarrow \text{grad } Q = 0 \end{cases}$

$\Rightarrow y_p = e^{2x}(S(x)\cos 0x + T(x)\sin 0x) = e^{2x}S(x)$

grad $S = 0$ $\Rightarrow y_p = c_3 e^{2x}$

Se inlocuieşte în ecuaţia iniţială $\Rightarrow 4c_3 e^{2x} + 2c_3 e^{2x} - 2c_3 e^{2x} = e^{2x} \; \forall x$

$\Rightarrow 4c_3 = 1 \iff c_3 = \frac{1}{4} \Rightarrow y_p = \frac{1}{4} e^{2x}$

$\Rightarrow y_0 = c_1 e^x + c_2 e^{-2x} + \frac{1}{4} e^{2x}$

b) $y'' - 4y' + 4y = \cos 2x$;

I. $y'' - 4y' + 4y = 0$

$\iff \lambda^2 - 4\lambda + 4 = 0 \iff (\lambda - 2)^2 = 0 \iff \lambda_{1,2} = 2$

$\Rightarrow e^{2x}, xe^{2x} \Rightarrow y_0 = c_1 e^{2x} + c_2 xe^{2x}$

II. $y'' - 4y' + 4y = \cos 2x$

$\cos 2x = e^{ax}(P(x)\cos(bx) + Q(x)\sin(bx)) \Rightarrow \begin{cases} a = 0 \\ b = 2 \\ Q(x) = 0 \\ P(x) = 1 \end{cases} \Rightarrow \max(\text{grad } P, \text{grad } Q) = 0$

$\Rightarrow y_p = e^{0x}(R(x)\cos(2x) + S(x)\sin(2x))$

dar grad $R, S \leq 0 \Rightarrow \begin{matrix} R(x) = k \\ S(x) = b \end{matrix} \Rightarrow y_p = k\cos 2x + b\sin 2x$

$\Rightarrow y_p' = -2k\sin 2x + 2\ell\cos 2x \Rightarrow y_p'' = -4k\cos 2x - 4\ell\sin 2x$

$\Rightarrow$ înlocuim în ecuație:

$-4k\cos 2x - 4\ell\sin 2x - 4(2k\sin 2x + 2\ell\cos 2x) + 4(k\cos 2x + \ell\sin 2x) = \cos 2x$

$\Leftrightarrow -4k\cos 2x - 4\ell\sin 2x - 8k\sin 2x - 8\ell\cos 2x + 4k\cos 2x + 4\ell\sin 2x = \cos 2x$

$\Rightarrow \begin{cases} k = 0 \\ \ell = -\frac{1}{8} \end{cases} \Rightarrow y_p = -\frac{1}{8}\sin 2x$

$\Rightarrow y = y_0 + y_p = c_1 e^x + c_2 x e^x - \frac{1}{8}\sin 2x$

c) $y'' + y = \cos 2x + \sin 2x;$

I. $y'' + y = 0$

$\Leftrightarrow r^2 + 1 = 0 \Leftrightarrow r = \pm i \Rightarrow \alpha = 0, \beta = 1$

$e^0\cos x = \cos x$

$e^0\sin x = \sin x$

$\Rightarrow y_0 = c_1\cos x + c_2\sin x$

II. $y'' + y = \cos 2x + \sin 2x$

$\cos 2x + \sin 2x = e^{ax}(P(x)\cos(bx) + Q(x)\sin(bx)) \Rightarrow \begin{cases} a = 0 \\ P(x) = Q(x) = 1 \\ b = 2 \end{cases}$

$\Rightarrow y_p = R(x)\cos 2x + S(x)\sin 2x$

$\text{grad } R, \text{grad } S \le \max(\text{grad } P, \text{grad } Q) = 1 \Rightarrow \begin{matrix} R(x) = k \\ S(x) = \ell \end{matrix}$

$\Rightarrow y_p = k\cos 2x + \ell\sin 2x$

$\Rightarrow y_p' = -2k\sin 2x + 2\ell\cos 2x \Rightarrow y_p'' = -4k\cos 2x - 4\ell\sin 2x$

$\Rightarrow y_p'' + y_p = \cos 2x + \sin 2x \Leftrightarrow -4k\cos 2x - 4\ell\sin 2x + k\cos 2x + \ell\sin 2x = \cos 2x + \sin 2x \quad \forall x$

$\Leftrightarrow \begin{cases} -3k = 1 \\ -3\ell = 1 \end{cases} \Leftrightarrow k = \ell = -\frac{1}{3} \Rightarrow y_p = -\frac{1}{3}\cos 2x - \frac{1}{3}\sin 2x$

$\Rightarrow y = y_0 + y_p = c_1\cos x + c_2\sin x - \frac{1}{3}\cos 2x - \frac{1}{3}\sin 2x$

d) $y'' - 2y' + y = e^x + e^{-x}$;

I. $y'' - 2y' + y = 0$

$\Rightarrow \lambda^2 - 2\lambda + 1 = 0 \Leftrightarrow (\lambda - 1)^2 = 0 \Leftrightarrow \lambda_{1,2} = 1$

$\Rightarrow e^x, xe^x \Rightarrow y_0 = c_1 e^x + c_2 x e^x$

II. $y'' - 2y' + y = e^x$

$e^x = e^{ax}\left(P(x)\cos(bx) + Q(x)\sin(bx)\right) = pe^x \Rightarrow y''_{p1} = y'_{p1} = pe^x$

$\Rightarrow pe^x - 2pe^x + pe^x = e^x \Leftrightarrow 0 = e^x$ fals $\Rightarrow \nexists\, y_{p1}$

III. $y'' - 2y' + y = e^{-x}$

$\Rightarrow e^{-x} = pe^{-x} \Rightarrow y'_{p2} = -pe^{-x} \Rightarrow y''_{p2} = pe^{-x}$

$\Rightarrow pe^{-x} - 2(-pe^{-x}) + pe^{-x} = e^{-x} \Leftrightarrow 4pe^{-x} = e^{-x} \ \forall x \Leftrightarrow p = \frac{1}{4}$

$\Rightarrow y_{p2} = \frac{1}{4} e^{-x}$

$\Rightarrow y = y_0 + y_{p2} = c_1 e^x + c_2 x e^x + \frac{1}{4} e^{-x}$

e) $y'' - 2y' + y = e^x \sin x$;

I. $y'' - 2y' + y = 0$

$\lambda^2 - 2\lambda + 1 = 0 \Leftrightarrow (\lambda - 1)^2 = 0 \Rightarrow \lambda_{1,2} = 1$

$\Rightarrow e^x, xe^x \Rightarrow y_0 = c_1 e^x + c_2 x e^x$

II. $y'' - 2y' + y = e^x \sin x$

$e^x \sin x = e^{ax}\left(P(x)\cos(bx) + Q(x)\sin(bx)\right) \Rightarrow \begin{cases} a = 1 \\ b = 1 \\ P(x) = 0 \\ Q(x) = 1 \end{cases}$

$\Rightarrow y_p = e^x\left(R(x)\cos x + S(x)\sin x\right)$

$\max(\text{grad } R, \text{grad } S) \le \max(\text{grad } P, \text{grad } Q) = 1 \Rightarrow R(x) = r,\ S(x) = s$

$\Rightarrow y_p = e^x(r\cos x + s\sin x) \Rightarrow y'_p = e^x(r\cos x + s\sin x) + e^x(-r\sin x + s\cos x)$

$= e^x((r+s)\cos x + (s-r)\sin x)$

$\Rightarrow y_p'' = e^x((h+s)\cos x + (s-h)\sin x) + e^x(-(h+s)\sin x + (s-h)\cos x)$

$\Rightarrow e^x((h+s)\cos x + (s-h)\sin x) + e^x(-(h+s)\sin x + (s-h)\cos x) -$

$-2e^x((h+s)\cos x + (s-h)\sin x) + e^x(h\cos x + s\sin x) = e^x \sin x \mid : e^x \neq 0$

$\Rightarrow$ ~~$h\cos x + s\cos x + s\sin x - h\sin x = h\sin x - s\sin x + s\cos x - h\cos x$~~ $= 2h\cos x - 2s\cos x$

$-2s\sin x + 2h\sin x + h\cos x + s\sin x = \sin x$

$\Leftrightarrow -h\cos x - s\sin x = \sin x \Rightarrow \begin{cases} h = 0 \\ s = -1 \end{cases} \Rightarrow y_p = -e^x \sin x$

$\Rightarrow y = y_0 + y_p = c_1 e^x + c_2 x e^x - e^x \sin x$

Toate se fac la fel ...

30. Să se rezolve, cu ajutorul matricei exponenţiale, următoarele sisteme de ecuaţii diferenţiale, ştiind că $x_1(0) = 0$ şi $x_2(0) = 1$:

a) $\begin{cases} x_1' = -x_1 + 4x_2 + e^t \\ x_2' = x_1 - x_2 + 1 \end{cases}$;

Not $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $A = \begin{pmatrix} -1 & 4 \\ 1 & -1 \end{pmatrix}$, $b = \begin{pmatrix} e^t \\ 1 \end{pmatrix}$

$\Rightarrow X' = AX + b$

$$X(t) = e^{tA} \cdot X(0) + e^{tA} \int_0^t e^{-sA} b(s)\, ds$$

$x_1(0) = 0$
$x_2(0) = 1$ $\Rightarrow X(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\det(A - \lambda I_2) = \begin{vmatrix} -1-\lambda & 4 \\ 1 & -1-\lambda \end{vmatrix} = 0 \Leftrightarrow (1+\lambda)^2 - 2^2 = 0 \Leftrightarrow (\lambda - 1)(3+\lambda) = 0$

$\Rightarrow \lambda_1 = -3, \lambda_2 = 1$

I. $AV = V \Leftrightarrow \begin{cases} -v_1 + 4v_2 = v_1 \\ v_1 - v_2 = v_2 \end{cases} \Leftrightarrow v_1 = 2v_2 \Rightarrow V_1 = sp\left\{\begin{pmatrix} 2 \\ 1 \end{pmatrix}\right\}$

$\left(= sp\left\{\begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}\right\} \right)$

II. $AV=-3V \iff \begin{cases} -v_1+4v_2=-3v_1 \\ v_1-v_2=-3v_2 \end{cases} \iff v_1=-2v_2 \Rightarrow V_{-3}=\operatorname{Sp}\left\{\begin{pmatrix}-2\\1\end{pmatrix}\right\}$

$\left(=\operatorname{Sp}\left\{\begin{pmatrix}\frac{-2}{\sqrt{5}}\\ \frac{1}{\sqrt{5}}\end{pmatrix}\right\}\right)$

$\Rightarrow P=\begin{pmatrix}2&-2\\1&1\end{pmatrix}, D=\begin{pmatrix}1&0\\0&-3\end{pmatrix} \Rightarrow A=PDP^{-1}$

$\Rightarrow A=\begin{pmatrix}2&-2\\1&1\end{pmatrix}\begin{pmatrix}1&0\\0&-3\end{pmatrix}\frac{1}{4}\begin{pmatrix}1&2\\-1&2\end{pmatrix} \Rightarrow e^{tA}=\begin{pmatrix}2&-2\\1&1\end{pmatrix}\begin{pmatrix}e^t&0\\0&e^{-3t}\end{pmatrix}\frac{1}{4}\begin{pmatrix}1&2\\-1&2\end{pmatrix}$

$=\frac{1}{4}\begin{pmatrix}2e^t&-2e^{-3t}\\e^t&e^{-3t}\end{pmatrix}\begin{pmatrix}1&2\\-1&2\end{pmatrix}=\frac{1}{4}\begin{pmatrix}2e^t+2e^{-3t}&4e^t-4e^{-3t}\\e^t-e^{-3t}&2e^t+2e^{-3t}\end{pmatrix}$

$\Rightarrow \int_0^t e^{-sA}b(s)\,ds=\frac{1}{4}\int_0^t\begin{pmatrix}2e^{-s}+2e^{3s}&4e^{-s}-4e^{3s}\\e^{-s}-e^{3s}&2e^{-s}+2e^{3s}\end{pmatrix}\begin{pmatrix}e^s\\1\end{pmatrix}ds=$

$=\frac{1}{4}\int_0^t\begin{pmatrix}(2e^{-s}+2e^{3s})e^s+4e^{-s}-4e^{3s}\\(e^{-s}-e^{3s})e^s+2e^{-s}+2e^{3s}\end{pmatrix}ds=\frac{1}{4}\int_0^t\begin{pmatrix}2+2e^{4s}+4e^{-s}-4e^{3s}\\1-e^{4s}+2e^{-s}+2e^{3s}\end{pmatrix}ds$

$=\frac{1}{4}\begin{pmatrix}\int_0^t(2+2e^{4s}+4e^{-s}-4e^{3s})ds\\ \int_0^t(1-e^{4s}+2e^{-s}+2e^{3s})ds\end{pmatrix}=\frac{1}{4}\begin{pmatrix}2s+\frac{2}{4}e^{4s}+\frac{4}{-1}e^{-s}-\frac{4}{3}e^{3s}\Big|_0^t\\ s-\frac{1}{4}e^{4s}+\frac{2}{-1}e^{-s}+\frac{2}{3}e^{3s}\Big|_0^t\end{pmatrix}$

$\left\{\int e^{as}ds=\frac{1}{a}\int a e^{as}ds=\frac{1}{a}\int e^x dx=\frac{1}{a}e^x=\frac{1}{a}e^{as}\right.$
$\left.\text{Not } as=x \iff a\,ds=dx\right\}$

$=\frac{1}{4}\begin{pmatrix}2s+\frac{1}{2}e^{2s}-4e^{-s}-\frac{4}{3}e^{3s}\Big|_0^t\\ s-\frac{1}{4}e^{2s}-2e^{-s}+\frac{2}{3}e^{3s}\Big|_0^t\end{pmatrix}=\frac{1}{4}\begin{pmatrix}2t+e^{2t}-4e^{-t}-\frac{4}{3}e^{3t}-\frac{55}{6}\\ t-\frac{1}{2}e^{2t}-2e^{-t}+\frac{2}{3}e^{3t}-\frac{19}{4}\end{pmatrix}$

$x(t)=e^{tA}\cdot x(0)+e^{tA}\int_0^t e^{-sA}b(s)\,ds=\dots\dots$

b) $\begin{cases} x_1' = 2x_1 + x_2 + t \\ x_2' = x_1 + 2x_2 + 1 + e^{-t} \end{cases}$;

$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $b = \begin{pmatrix} t \\ e^{-t} \end{pmatrix}$

$X(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$X(t) = e^{tA} \cdot X(0) + e^{tA} \int_0^t e^{-sA} \cdot b(s)\, ds$$

$\det(A - \lambda I_2) = 0 \Leftrightarrow \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0 \Leftrightarrow (2-\lambda)^2 - 1^2 = 0 \Leftrightarrow$
$\Leftrightarrow (1-\lambda)(3-\lambda) = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = 3$

I. $AV = V \Leftrightarrow \begin{cases} 2v_1 + v_2 = v_1 \\ v_1 + 2v_2 = v_2 \end{cases} \Leftrightarrow v_2 = -v_1 \Rightarrow V_1 = sp\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\}$

II. $AV = 3V \Leftrightarrow \begin{cases} 2v_1 + v_2 = 3v_1 \\ v_1 + 2v_2 = 3v_2 \end{cases} \Leftrightarrow v_1 = v_2 \Rightarrow V_3 = sp\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$

$\Rightarrow P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$ $\Rightarrow A = PDP^{-1}$

$\hookrightarrow P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

$$\Rightarrow e^{tA} = P \cdot e^{tD} \cdot P^{-1} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} e^t & 0 \\ 0 & e^{3t} \end{pmatrix}\frac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} e^t & e^{3t} \\ -e^t & e^{3t} \end{pmatrix}\frac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} =$$

$$= \frac{1}{2}\begin{pmatrix} e^t + e^{3t} & -e^t + e^{3t} \\ -e^t + e^{3t} & e^t + e^{3t} \end{pmatrix}$$

$$\Rightarrow \frac{1}{2}\int_0^t e^{-sA} b(s)\, ds = \frac{1}{2}\int_0^t \begin{pmatrix} e^{-s} + e^{-3s} & -e^{-s} + e^{-3s} \\ -e^{-s} + e^{-3s} & e^{-s} + e^{-3s} \end{pmatrix}\begin{pmatrix} s \\ e^{-s} \end{pmatrix} ds \quad \ldots \ldots$$

Se înlocuiesc apoi în $f: X(t) = e^{tA} X(0) + e^{tA} \int_0^t e^{-sA} b(s)\, ds$
până se ajunge la $X(t) = \begin{pmatrix} ceva\ 1 \\ ceva\ 2 \end{pmatrix} \Rightarrow \begin{matrix} x_1(t) = ceva\ 1 \\ x_2(t) = ceva\ 2 \end{matrix}$

c) $\begin{cases} x_1' = x_1 + 4x_2 + \sin t \\ x_2' = 2x_1 + 3x_2 + \cos 2t \end{cases}$;

$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$, $b = \begin{pmatrix} \sin t \\ \cos 2t \end{pmatrix}$

$X' = AX + b$, $X(0) = \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$x(t) = e^{tA}\cdot x(0) + e^{tA}\int_0^t e^{-sA} b(s)\,ds$$

$$\det(A-\lambda I_2)=0 \Leftrightarrow \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0 \Leftrightarrow (1-\lambda)(3-\lambda)-8=0 \Leftrightarrow 3-\lambda-3\lambda+\lambda^2-8=0$$

$$\Leftrightarrow \lambda^2-4\lambda-5=0 \Leftrightarrow \lambda^2-5\lambda+\lambda-5=0 \Leftrightarrow (\lambda+1)(\lambda-5)=0$$

$$\Rightarrow \lambda_1=-1,\ \lambda_2=5$$

$$\text{I. } AV=-V \Leftrightarrow \begin{cases} v_1+4v_2=-v_1 \\ 2v_1+3v_2=-v_2 \end{cases} \Leftrightarrow v_1=-2v_1 \Rightarrow V_{-1}=\mathrm{Sp}\left\{\begin{pmatrix}-2\\1\end{pmatrix}\right\}$$

$$\text{II. } AV=5V \Leftrightarrow \begin{cases} v_1+4v_2=5v_1 \\ 2v_1+3v_2=5v_2 \end{cases} \Leftrightarrow v_1=v_2 \Rightarrow V_{-1}=\mathrm{Sp}\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\}$$

$$\Rightarrow P=\begin{pmatrix}-2 & 1\\ 1 & 1\end{pmatrix},\ D=\begin{pmatrix}-1 & 0\\ 0 & 5\end{pmatrix} \Rightarrow A=PDP^{-1}$$

$$\hookrightarrow P^{-1}=-\frac{1}{3}\begin{pmatrix}1 & -1\\ -1 & -2\end{pmatrix}$$

$$\Rightarrow e^{tA}=Pe^{tD}P^{-1}=\begin{pmatrix}-2 & 1\\ 1 & 1\end{pmatrix}\begin{pmatrix}e^{-t} & 0\\ 0 & e^{5t}\end{pmatrix}\left(-\frac{1}{3}\right)\begin{pmatrix}1 & -1\\ -1 & -2\end{pmatrix}=$$

$$=-\frac{1}{3}\begin{pmatrix}-2e^{-t} & e^{5t}\\ e^{-t} & e^{5t}\end{pmatrix}\begin{pmatrix}1 & -1\\ -1 & -2\end{pmatrix}=-\frac{1}{3}\begin{pmatrix}-2e^{-t}-e^{5t} & 2e^{-t}-2e^{5t}\\ e^{-t}-e^{5t} & -e^{-t}-2e^{5t}\end{pmatrix}$$

$$\int_0^t e^{-sA} b(s)\,ds=-\frac{1}{3}\int_0^t \begin{pmatrix}-2e^{s}-e^{-5s} & 2e^{s}-2e^{-5s}\\ e^{s}-e^{-5s} & -e^{s}-2e^{-5s}\end{pmatrix}\begin{pmatrix}\sin s\\ \cos 2s\end{pmatrix} ds \quad \ldots$$

Se înlocuiește în f. $x(t)=e^{tA}\cdot x(0)+e^{tA}\int_0^t e^{-sA}b(s)\,ds$

.... nu mai fac restul subpunctelor că urlu

SUBIECT DRIVE

Numele şi prenumele: _____
Anul I, Seria B, Grupa: _____
Disciplina: Algebră liniară, geometrie analitică şi ecuaţii diferenţiale
Data: 10 februarie 2023

EXAMENUL DIN SESIUNE

Problema 1. a) Să se justifice şi, apoi, să se găsească descompunerea QR a matricei $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 1 \end{pmatrix}$.

b) Folosind a), să se determine soluţia în sensul celor mai mici pătrate a sistemului $\begin{cases} x_1 + 2x_2 = 1, \\ 2x_1 + x_2 = 1, \\ 3x_1 + x_2 = 1. \end{cases}$

Problema 2. Se consideră matricea $A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$. Să se determine:

i) valorile proprii ale matricei A şi câte o bază pentru subspaţiile proprii corespunzătoare acestora;
ii) o matrice ortogonală, $Q \in \mathcal{M}_3(\mathbb{R})$, şi o matrice diagonală, $D \in \mathcal{M}_3(\mathbb{R})$, astfel încât $A = QDQ^T$;
iii) matricea A^n, unde $n \in \mathbb{N}^*$.

Problema 3. Folosind metoda transformărilor ortogonale, să se aducă la forma canonică următoarea conică: $13x_1^2 + 10x_1x_2 + 13x_2^2 - 36x_1 - 36x_2 - 36 = 0$.

Problema 4. Să se rezolve ecuaţiile diferenţiale:
a) $xy' = 2x^2\sqrt{y} + 4y$; b) $3x\left(x + 2y^2\right)\mathrm{d}x + 2y\left(3x^2 + 2y^2\right)\mathrm{d}y = 0$.

Problema 5. Să se rezolve cu ajutorul matricei exponenţiale următorul sistem de ecuaţii diferenţiale:
$\begin{cases} x_1'(t) = 5x_1(t) - 3x_2(t), \\ x_2'(t) = -6x_1(t) + 2x_2(t) + e^t, \end{cases}$ ştiind că $x_1(0) = 0$ şi $x_2(0) = 1$.

1.

a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 1 \end{pmatrix}$ $\quad u_1 \quad u_2$

$v_1 = u_1$

$v_2 = u_2 - \text{pr}_{v_1} u_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = u_2 - \frac{7}{14}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$= \begin{pmatrix} \frac{3}{2} \\ 0 \\ -\frac{1}{2} \end{pmatrix}$

$\Rightarrow w_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{14}}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$w_2 = \frac{v_2}{\|v_2\|} = \frac{2}{\sqrt{10}}\begin{pmatrix} \frac{3}{2} \\ 0 \\ -\frac{1}{2} \end{pmatrix} = \frac{1}{\sqrt{10}}\begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}$

$\Rightarrow Q = \begin{pmatrix} \frac{1}{\sqrt{14}} & \frac{3}{\sqrt{10}} \\ \frac{2}{\sqrt{14}} & 0 \\ \frac{3}{\sqrt{14}} & -\frac{1}{\sqrt{10}} \end{pmatrix}$

$$R = \begin{pmatrix} \langle w_1, u_1 \rangle & \langle w_1, u_2 \rangle \\ 0 & \langle w_2, u_2 \rangle \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{14}} \cdot 14 & \frac{4}{\sqrt{14}} \\ 0 & \frac{5}{\sqrt{10}} \end{pmatrix}$$

$$= \begin{pmatrix} \sqrt{14} & \frac{\sqrt{14}}{2} \\ 0 & \frac{\sqrt{10}}{2} \end{pmatrix}$$

$\Rightarrow A = QR$

b) $\begin{cases} x_1 + 2x_2 = 1 \\ 2x_1 + x_2 = 1 \\ 3x_1 + x_2 = 1 \end{cases}$ $\quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$\Rightarrow \boxed{R\tilde{x} = Q^t b}$, $x = \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} \Rightarrow \begin{pmatrix} \sqrt{14} & \frac{\sqrt{14}}{2} \\ 0 & \frac{\sqrt{10}}{2} \end{pmatrix} \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{14}} & \frac{2}{\sqrt{14}} & \frac{3}{\sqrt{14}} \\ \frac{3}{\sqrt{10}} & 0 & -\frac{1}{\sqrt{10}} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$$\Leftrightarrow \begin{cases} \sqrt{14}\,\tilde{x}_1 + \frac{\sqrt{14}}{2}\tilde{x}_2 = \frac{6}{\sqrt{14}} \\ \frac{\sqrt{10}}{2}\tilde{x}_2 = \frac{2}{\sqrt{10}} \quad | \cdot 2\sqrt{10} \end{cases} \Leftrightarrow 10\tilde{x}_2 = 4 \Leftrightarrow \tilde{x}_2 = \frac{4}{10}$$

$$\sqrt{14}\,\tilde{x}_1 + \frac{\sqrt{14}}{2}\tilde{x}_2 = \frac{6}{\sqrt{14}} \quad | \cdot 2\sqrt{14} \Leftrightarrow 28\tilde{x}_1 + 14\tilde{x}_2 = 12$$

$$\Leftrightarrow 28\tilde{x}_1 + \cancel{14}^{7} \cdot \frac{4}{\cancel{10}_{5}} = 12$$

$$\Leftrightarrow 28\tilde{x}_1 + \frac{28}{5} = 12 \Leftrightarrow 28\tilde{x}_1 = \frac{32}{5} \quad | : 4$$

$$\Leftrightarrow \tilde{x}_1 = \frac{8}{35}$$

2. $A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$

a) $\det(A - \lambda I_3) = 0 \Leftrightarrow \begin{vmatrix} 4-\lambda & 2 & 2 \\ 2 & 4-\lambda & 2 \\ 2 & 2 & 4-\lambda \end{vmatrix} = (4-\lambda)^3 + 8 + 8 - 12(4-\lambda) = 0$

$= (4-\lambda)(16 - 8\lambda + \lambda^2 - 12) + 16 = 0$

$\Leftrightarrow (4-\lambda)(\lambda^2 - 8\lambda + 4) + 16 = 0 \Leftrightarrow 4\lambda^2 - 32\lambda + 16 - \lambda^3 + 8\lambda^2 - 4\lambda + 16 = 0$

$\Leftrightarrow -\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0$

Se observă că $\lambda = 2 \in Q \Rightarrow$

	λ^3	λ^2	λ^1	λ^0
	-1	12	-36	32
2	-1	10	-16	0

$\Leftrightarrow (\lambda-2)(-\lambda^2+10\lambda-16)=0 \Rightarrow \lambda_1=2$

$\Downarrow$ $\lambda^2-8\lambda-2\lambda+16=0 \Leftrightarrow (\lambda-2)(\lambda-8)=0$

$\Rightarrow \lambda_2=\lambda_1=2$

$\lambda_3=8$

I. $AV=2V$

$\Leftrightarrow \begin{pmatrix}4&2&2\\2&4&2\\2&2&4\end{pmatrix}\begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix}=2\begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix} \Leftrightarrow v_1+v_2+v_3=0 \Rightarrow v_2=-v_1-v_3$

$\Rightarrow V_2=\mathrm{sp}\left\{\begin{pmatrix}1\\-1\\0\end{pmatrix},\begin{pmatrix}0\\-1\\1\end{pmatrix}\right\}$

II. $AV=8V$

$\Leftrightarrow \begin{pmatrix}4&2&2\\2&4&2\\2&2&4\end{pmatrix}\begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix}=8\begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix} \Leftrightarrow \begin{cases}v_2+v_3=2v_1\\v_1+v_3=2v_2\\v_1+v_2=2v_3\end{cases} \Rightarrow 2v_3-v_2=2v_2-v_3$

$\Leftrightarrow v_2=v_3$

$\Rightarrow v_1=v_2=v_3$

$\Rightarrow V_8=\mathrm{sp}\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}\right\}$

b) $v_1=\begin{pmatrix}1\\-1\\0\end{pmatrix} \Leftrightarrow w_1=\frac{v_1}{\|v_1\|}=\begin{pmatrix}\frac{1}{\sqrt{2}}\\-\frac{1}{\sqrt{2}}\\0\end{pmatrix}$

$w_2=\frac{v_2}{\|v_2\|}=\begin{pmatrix}0\\-\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}$

$w_3=\frac{v_3}{\|v_3\|}=\begin{pmatrix}\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{pmatrix}$

$\Rightarrow Q=\begin{pmatrix}\frac{1}{\sqrt{2}}&0&\frac{1}{\sqrt{3}}\\-\frac{1}{\sqrt{2}}&-\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{3}}\\0&\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{3}}\end{pmatrix}$

$D=\begin{pmatrix}2&0&0\\0&2&0\\0&0&8\end{pmatrix}$

$\Rightarrow A=QDQ^t$

c) $A^n=(QDQ^t)^n=QDQ^tQDQ^t\cdot\ldots\cdot QDQ^t$

dar Q matrice ortogonală $\Rightarrow Q^tQ=I_n \Rightarrow A^n=QD^nQ^t$

$\Leftrightarrow A=\begin{pmatrix}\frac{1}{\sqrt{2}}&0&\frac{1}{\sqrt{3}}\\-\frac{1}{\sqrt{2}}&-\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{3}}\\0&\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{3}}\end{pmatrix}\begin{pmatrix}2^n&0&0\\0&2^n&0\\0&0&8^n\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{2}}&-\frac{1}{\sqrt{2}}&0\\0&-\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{3}}\end{pmatrix}=\ldots$ calcule

3. $g(x_1,x_2) = 13x_1^2 + 10x_1x_2 + 13x^2 - 36x_1 - 36x_2 - 36 = 0$

$\Leftrightarrow 13x_1^2 + 10x_1x_2 + 13x^2 - 36(x_1+x_2) - 36 = 0$

x^TAx, $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $A = \begin{pmatrix} 13 & 5 \\ 5 & 13 \end{pmatrix}$

$\det(A - \lambda I_2) = 0 \Leftrightarrow \begin{vmatrix} 13-\lambda & 5 \\ 5 & 13-\lambda \end{vmatrix} = 0 \Leftrightarrow (13-\lambda)^2 - 5^2 = 0 \Leftrightarrow (8-\lambda)(18-\lambda) = 0$

$\Rightarrow \lambda_1 = 8$

$\lambda_2 = 18$

I. $AV = 8V \Leftrightarrow \begin{cases} 13v_1 + 5v_2 = 8v_1 \\ 5v_1 + 13v_2 = 8v_2 \end{cases} \Leftrightarrow v_2 = -v_1 \Rightarrow V_8 = Sp\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\} = Sp\left\{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}\right\}$

II. $AV = 18V \Leftrightarrow \begin{cases} 13v_1 + 5v_2 = 18v_1 \\ 5v_1 + 13v_2 = 18v_2 \end{cases} \Leftrightarrow v_2 = v_1 \Rightarrow V_8 = Sp\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\} = Sp\left\{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}\right\}$

$\Rightarrow Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$, $D = \begin{pmatrix} 8 & 0 \\ 0 & 18 \end{pmatrix} \Rightarrow A = QDQ^t$

! mat notatie

$\Rightarrow x = Qy \Leftrightarrow \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \Leftrightarrow \begin{cases} x_1 = \frac{y_1}{\sqrt{2}} + \frac{y_2}{\sqrt{2}} = \frac{1}{\sqrt{2}}(y_1+y_2) \\ x_2 = -\frac{y_1}{\sqrt{2}} + \frac{y_2}{\sqrt{2}} = \frac{1}{\sqrt{2}}(-y_1+y_2) \end{cases}$

$\Rightarrow$ înlocuim în relația originală

$\tilde{g}(y_1,y_2) = 13 \cdot \frac{1}{2}(y_1+y_2)^2 + \frac{10}{2}(y_2^2 - y_1^2) + 13 \cdot \frac{1}{2}(-y_1+y_2)^2 - 36 \cdot \sqrt{2}y_2 - 36$

$= \frac{13}{2}y_1^2 + 13y_1y_2 + \frac{13}{2}y_2^2 + \frac{10}{2}y_2^2 - \frac{10}{2}y_1^2 + \frac{13}{2}y_1^2 - 13y_1y_2 + \frac{13}{2}y_2^2 - 36\sqrt{2}y_2 - 36$

$= \ldots$

4. a) $xy' = 2x^2\sqrt{y} + 4y$

$\Leftrightarrow y' = \frac{4}{x}y + 2xy^{\frac{1}{2}}$

$\Rightarrow \alpha = \frac{1}{2} \Rightarrow$ subst $u = y^{1-\alpha} = y^{\frac{1}{2}} = \sqrt{y} \Rightarrow y = u^2 \Leftrightarrow y' = 2uu'$

$\Leftrightarrow 2uu' = \frac{4}{x} \cdot u^2 + 2x \cdot u \quad | : 2u$

$\Leftrightarrow u' = \frac{2}{x}u + x$

u_0: $u' = \frac{2}{x}u \Leftrightarrow \frac{du}{u} = \frac{2}{x}dx \Leftrightarrow \ln|u| = 2\ln|x| + c$

$\Leftrightarrow u = \underline{\pm x^2 e^c} = k \quad = kx^2$

u_p: $u_p = k(x)x^2 \Rightarrow k'(x)x^2 + \cancel{k(x)\,2x} = \cancel{\frac{2}{x} \cdot k(x) \cdot x^2} + x$

$\Leftrightarrow k'(x) = \frac{1}{x} \Rightarrow k(x) = \ln|x| + \underbrace{c_1}_{=0} = \ln|x| \Rightarrow u_p = \ln|x|x^2$

$\Rightarrow u = u_0 + u_p = kx^2 + \ln|x| \cdot x^2 \Rightarrow y = u^2 = \left(kx^2 + \ln|x| \cdot x^2\right)^2$

b) $3x(x+2y^2)dx + 2y(3x^2+2y^2)dy = 0$

$\Leftrightarrow (6x^2y + 4y^3)dy = (3x^2 + 6xy^2)dx$

5. $\begin{cases} x_1'(t) = 5x_1(t) - 3x_2(t) \\ x_2'(t) = -6x_1(t) + 2x_2(t) + e^t \end{cases}$, $x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $A = \begin{pmatrix} 5 & -3 \\ -6 & 2 \end{pmatrix}$, $b = \begin{pmatrix} 0 \\ e^t \end{pmatrix}$

$\det(A - \lambda I_2) = 0 \Leftrightarrow \begin{vmatrix} 5-\lambda & -3 \\ -6 & 2-\lambda \end{vmatrix} = 0 \Leftrightarrow 10 - 7\lambda + \lambda^2 - 18 = 0$

$\Leftrightarrow \lambda^2 - 7\lambda - 8 = 0 \Leftrightarrow \lambda^2 - 8\lambda + \lambda - 8 = 0$

$\Leftrightarrow (\lambda+1)(\lambda-8) = 0 \Rightarrow \lambda_1 = -1$

$\lambda_2 = 8$

$\underline{I}.\ Av = -v \iff \begin{cases} 5v_1 - 3v_2 = -v_1 \\ -6v_1 + 2v_2 = -v_2 \end{cases} \Rightarrow v_2 = 2v_1 \Rightarrow V_{-1} = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\}$

$\underline{II}.\ Av = 8v \iff \begin{cases} 5v_1 - 3v_2 = 8v_1 \\ -6v_1 + 2v_2 = 8v_2 \end{cases} \iff v_2 = -v_1 \Rightarrow V_8 = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\}$

$\Rightarrow P = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix},\ D = \begin{pmatrix} -1 & 0 \\ 0 & 8 \end{pmatrix}$

$\hookrightarrow P^{-1} = -\frac{1}{3}\begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix}$

$\Rightarrow A = PDP^{-1} \Rightarrow e^{tA} = P\, e^{tD} P^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} (-1)^t & 0 \\ 0 & 8^t \end{pmatrix}\left(-\frac{1}{3}\right)\begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix}$

$= -\frac{1}{3}\begin{pmatrix} (-1)^t & 8^t \\ 2(-1)^t & -8^t \end{pmatrix}\begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix} = -\frac{1}{3}\begin{pmatrix} -(-1)^t - 2\cdot 8^t & -(-1)^t + 8^t \\ -2(-1)^t + 2\cdot 8^t & -2(-1)^t - 8^t \end{pmatrix}$

$$\boxed{X(t) = e^{tA}\cdot X(0) + e^{tA}\int_0^t e^{-sA} b(s)\,ds}$$

... a mai rămas să calculează